

DISC-NET ML Workshop

What Just Happened?



Architectures, Generative AI, LLMs

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- To today with accurate image classification

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- But how did it happen?

AlexNet

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- AlexNet was a reimplementation of CNNs first invented in the late 1990s by Yann LeCun
- What distinguishes AlexNet was it was implemented on two GPUs and trained on 1.3 million large images
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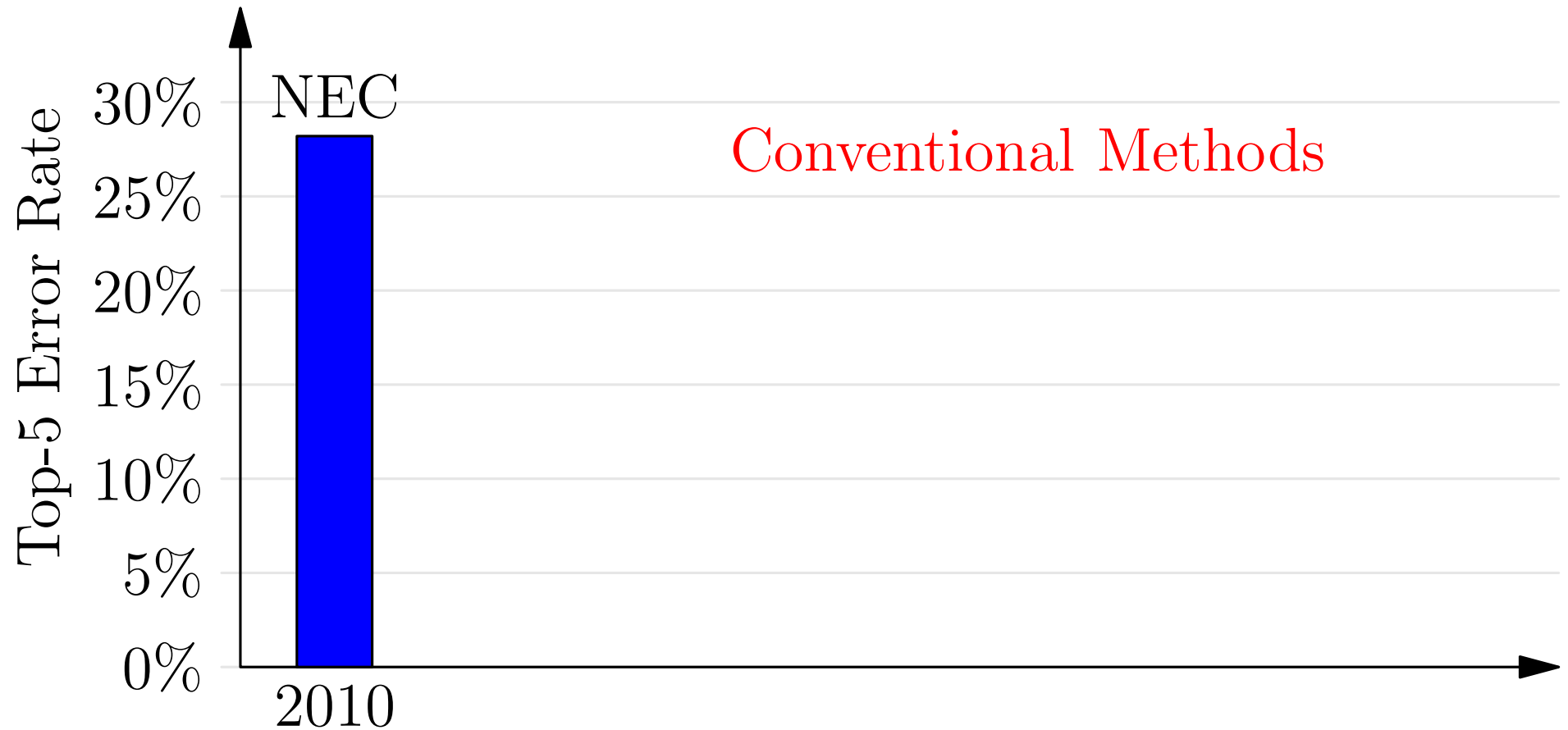
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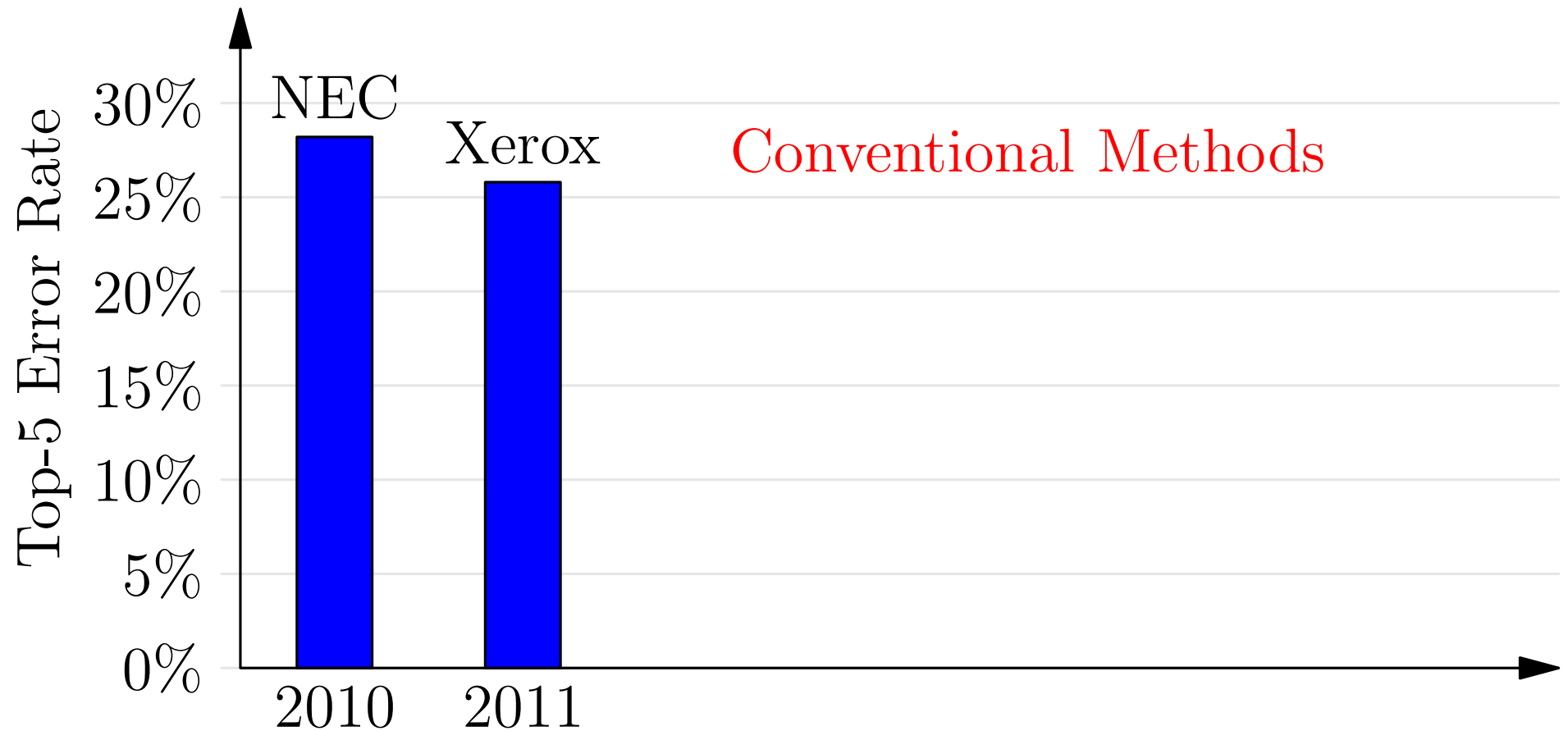
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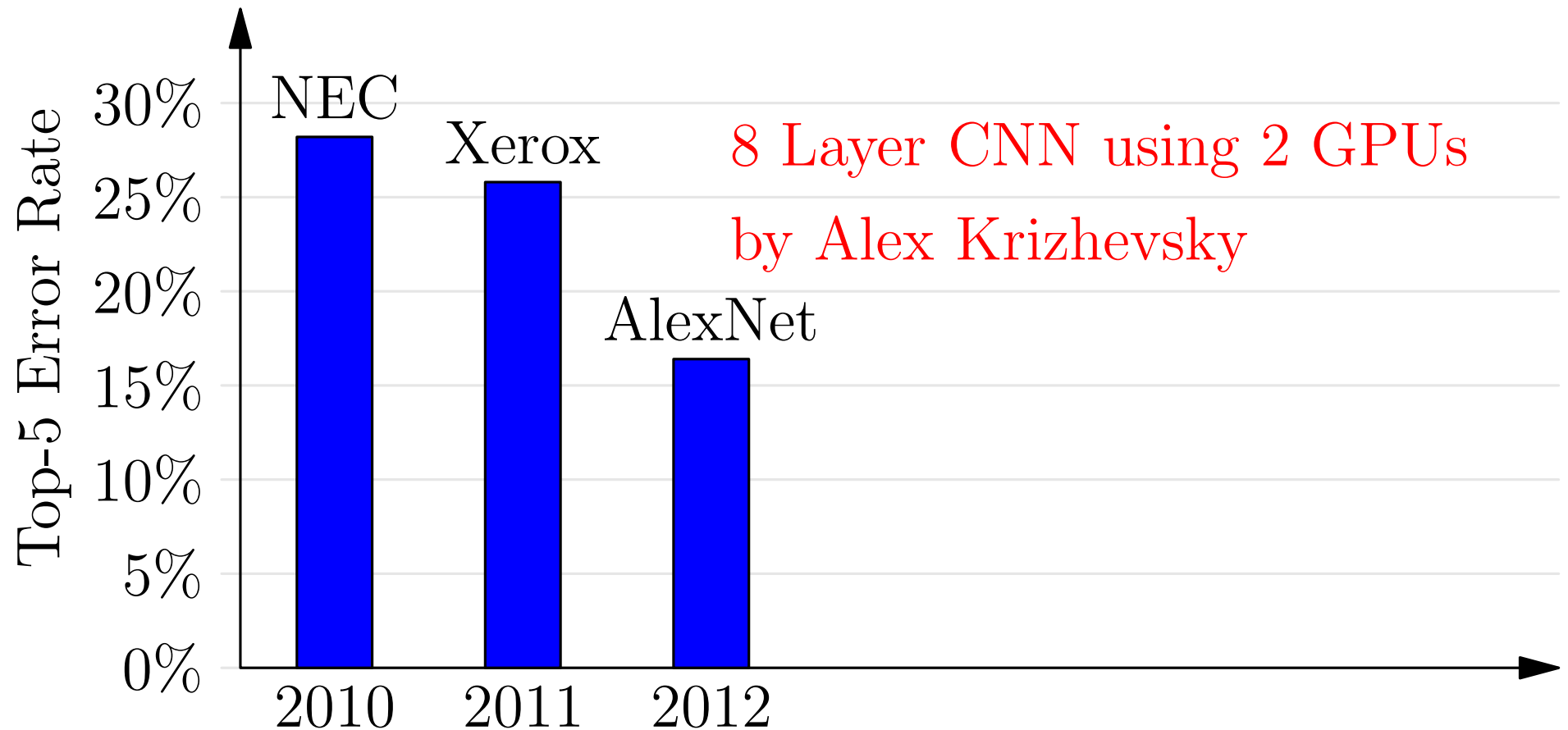
The Rapid Revolution



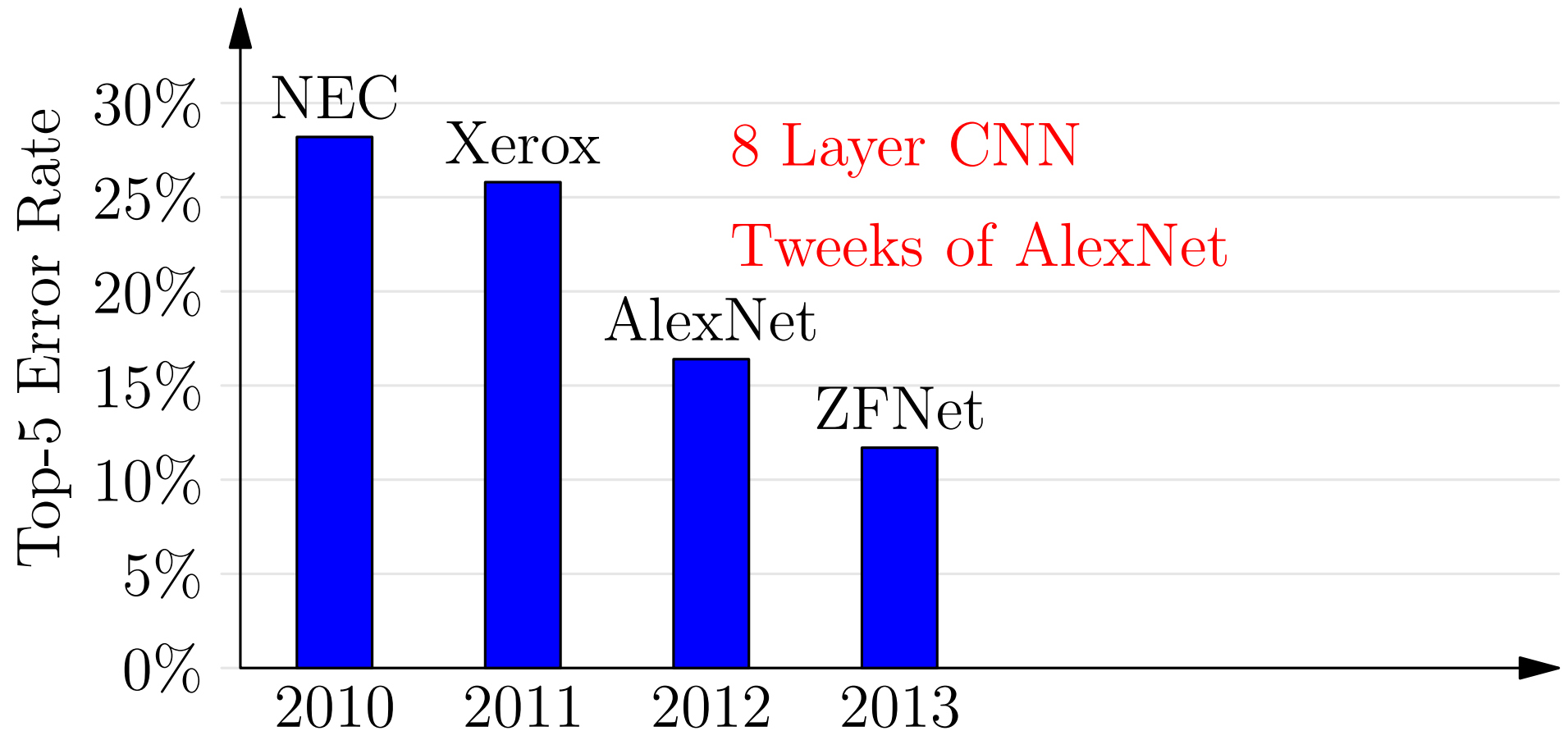
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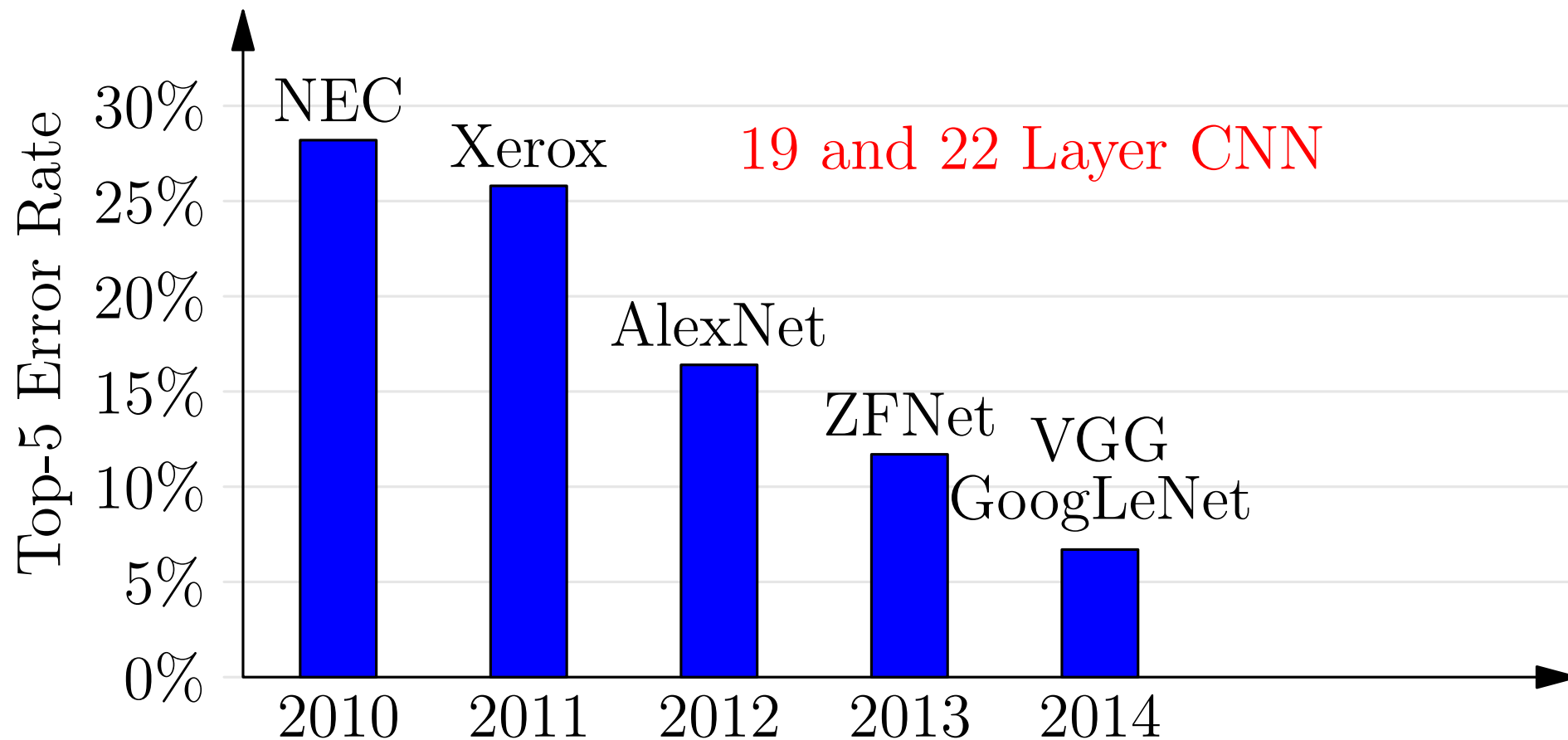
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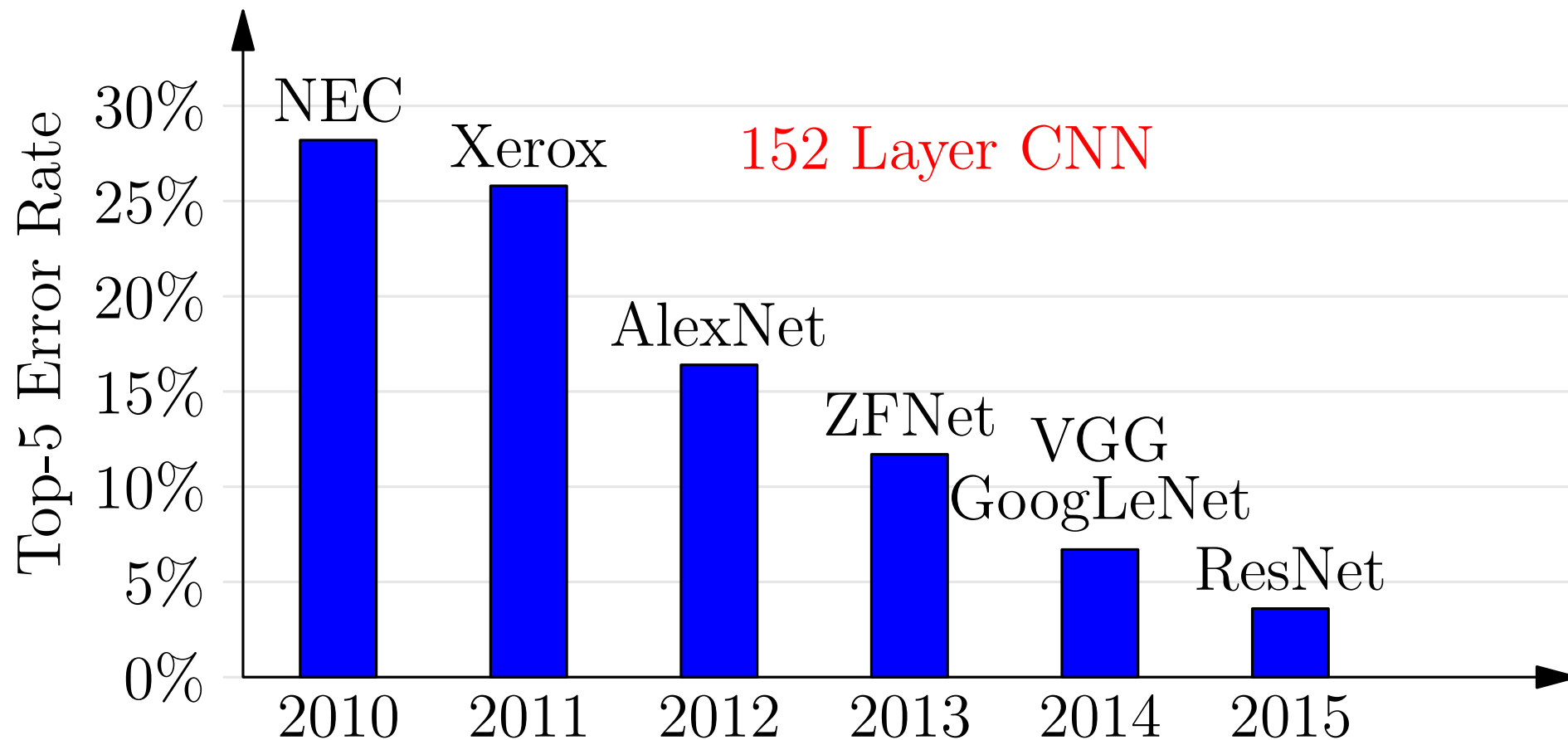
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Dataset Size Matters

- The surprising result was not only did the model do extremely on the ImageNet data
- But, the network could handle real world data
- If you train on a large enough dataset we discovered that machine learning actually worked in the real world!
- Better models would give small improvements in results, but data really matters!

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What was Different About AlexNet?

- The **CNN architecture** (although not new) was very well suited to vision (built-in translational equivariance)
- To train on bonkers amount of data you need a lot of compute, and AlexNet introduced **GPUs**
- Traditionally, we had to compute gradients by hand, making the design of large networks cumbersome
- New frameworks introduced **automatic differentiation** allowing users to create huge models without worrying about doing hard maths

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The Unreasonable Effectiveness of SGD

- An extraordinarily successful learning strategy is to design a standard network
- This scales well (linearly) to large networks and large datasets
- When you overfit, you just turn on some regularisation
- Sometimes gradient descent would struggle, but usually changing the optimiser to ADAM or fine-tuning some hyper-parameters would do the trick

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Computational Frameworks

- The Machine Learning community is rubbish when it comes to organising conferences and journals
- But when it comes to developing computational frameworks it is brilliant
- It rapidly developed, torch (Lua), TensorFlow, Keras, Pytorch, JAX, etc.
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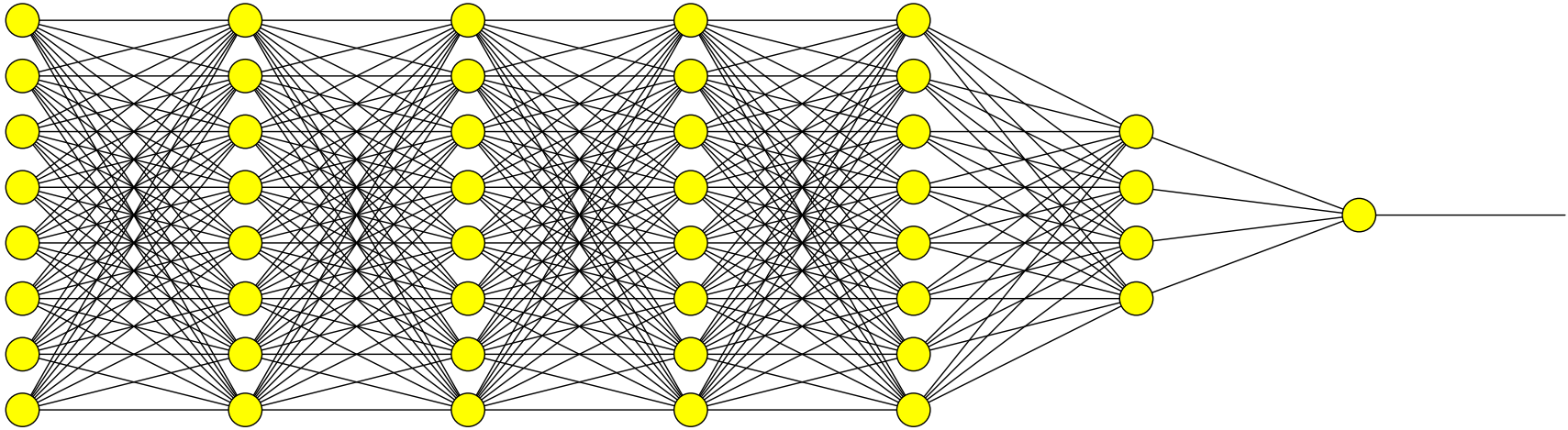
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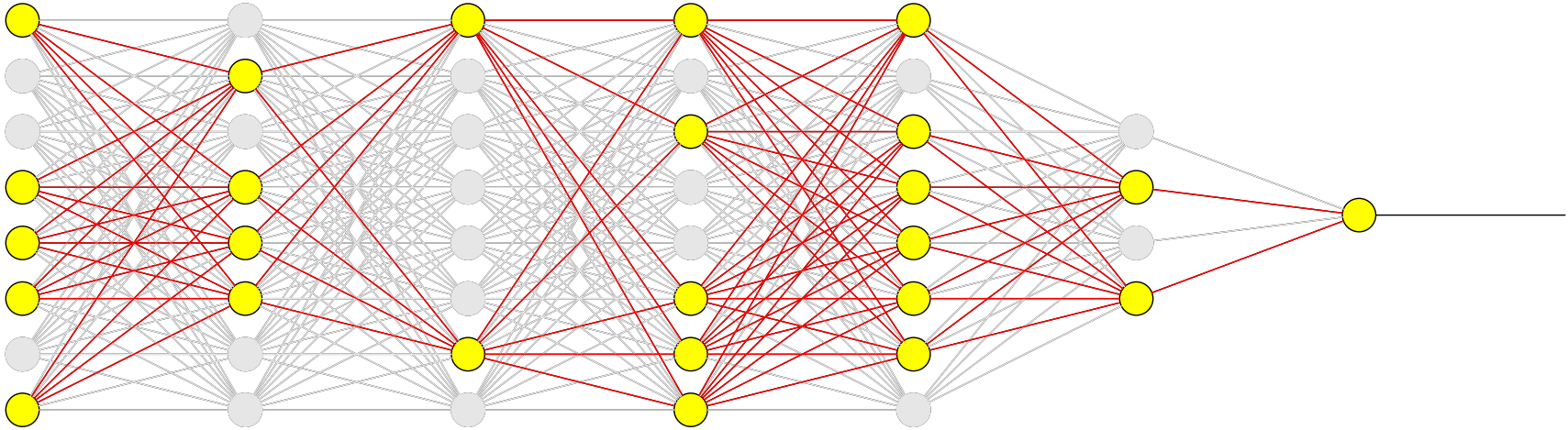
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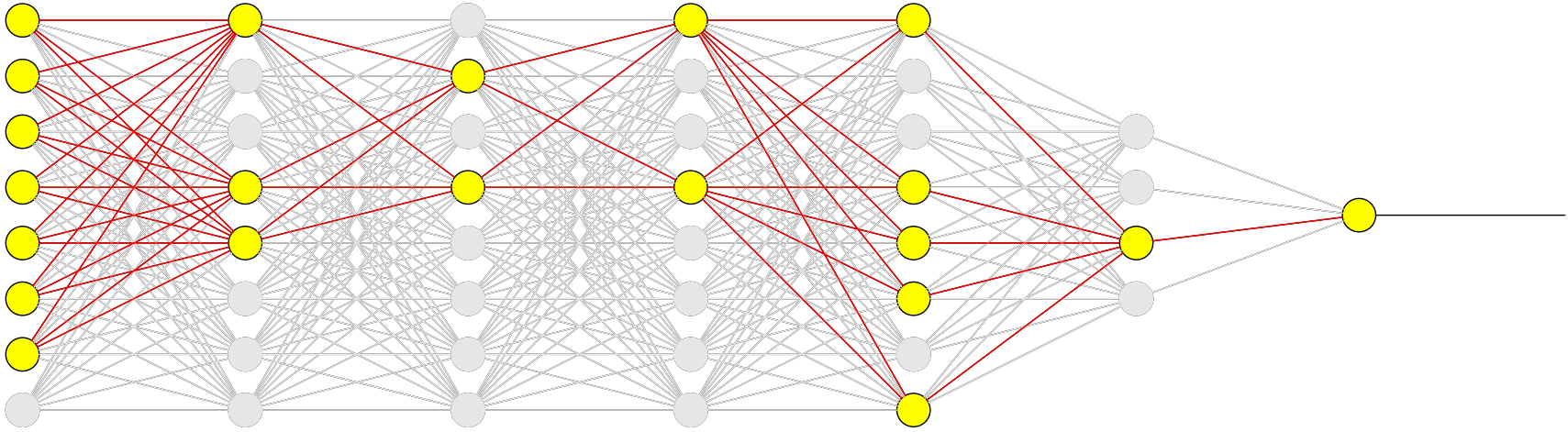
New Ideas: Dropout



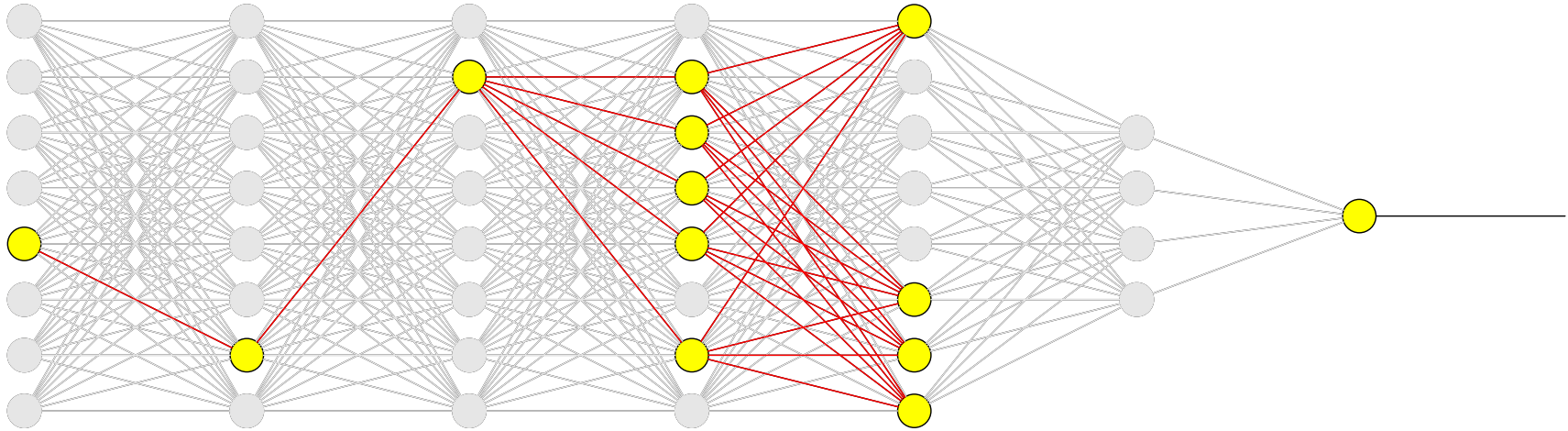
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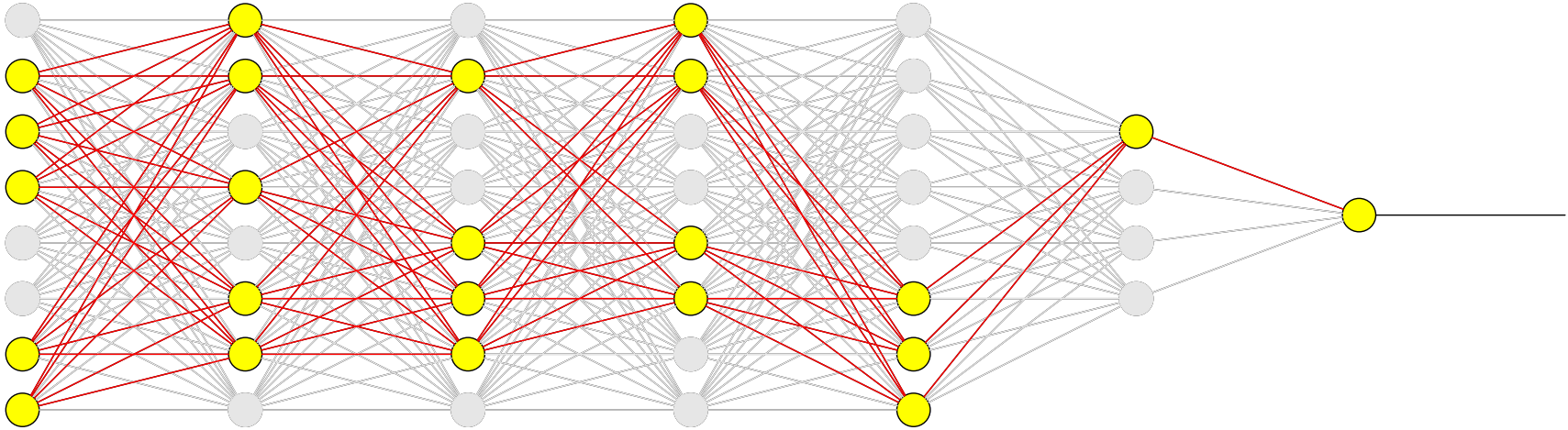
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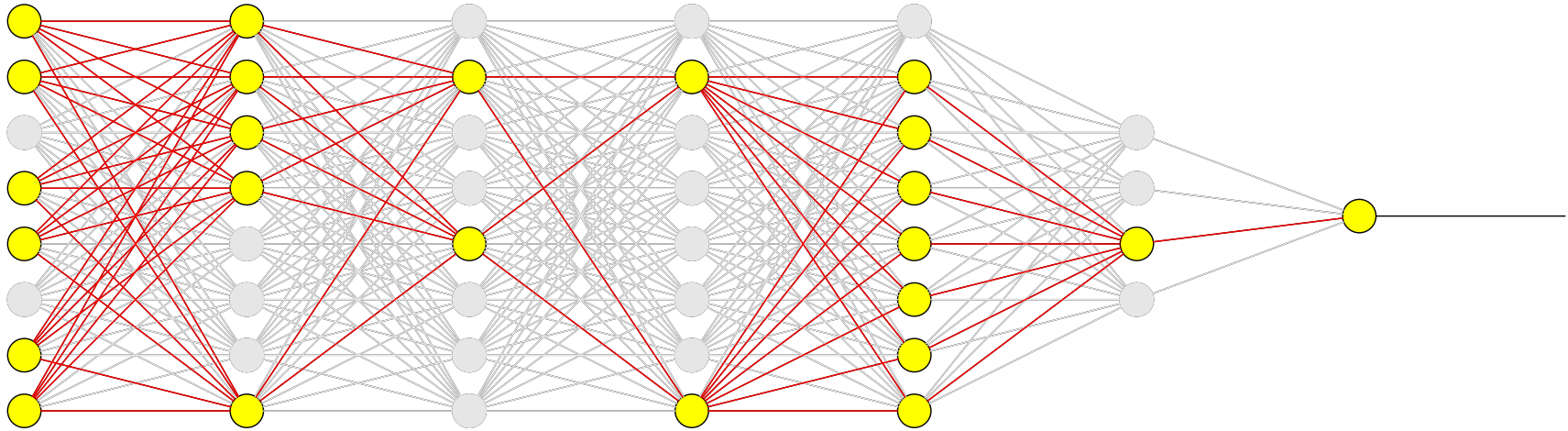
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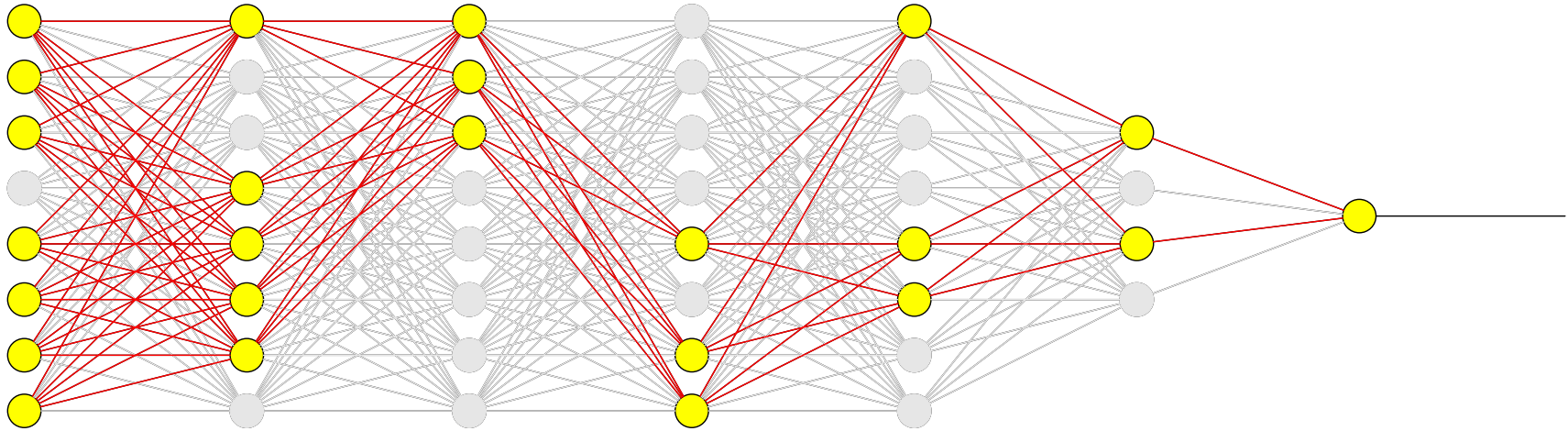
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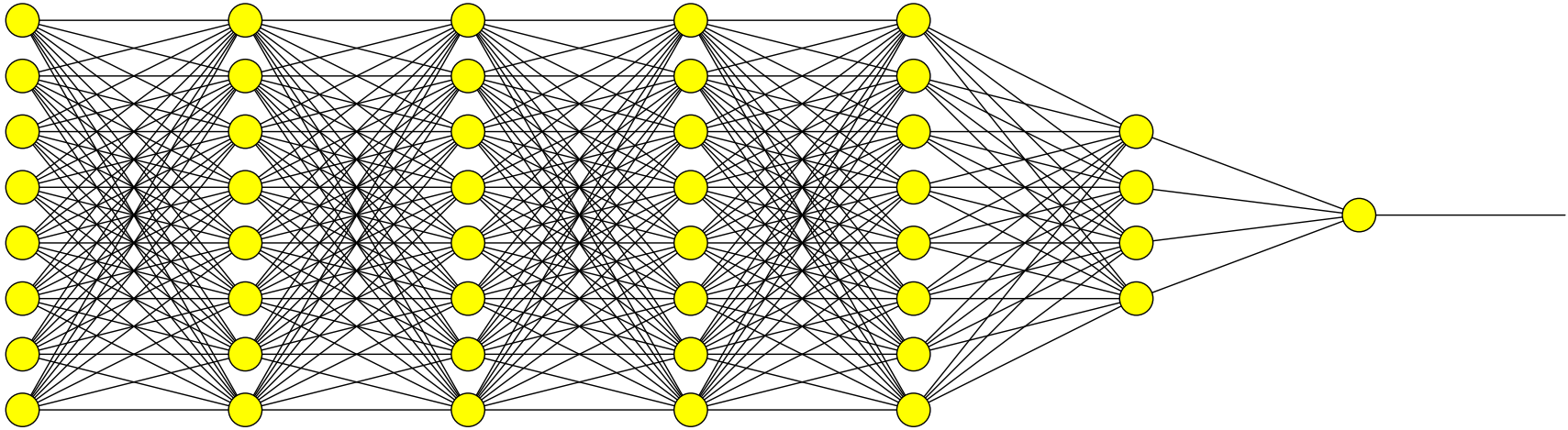
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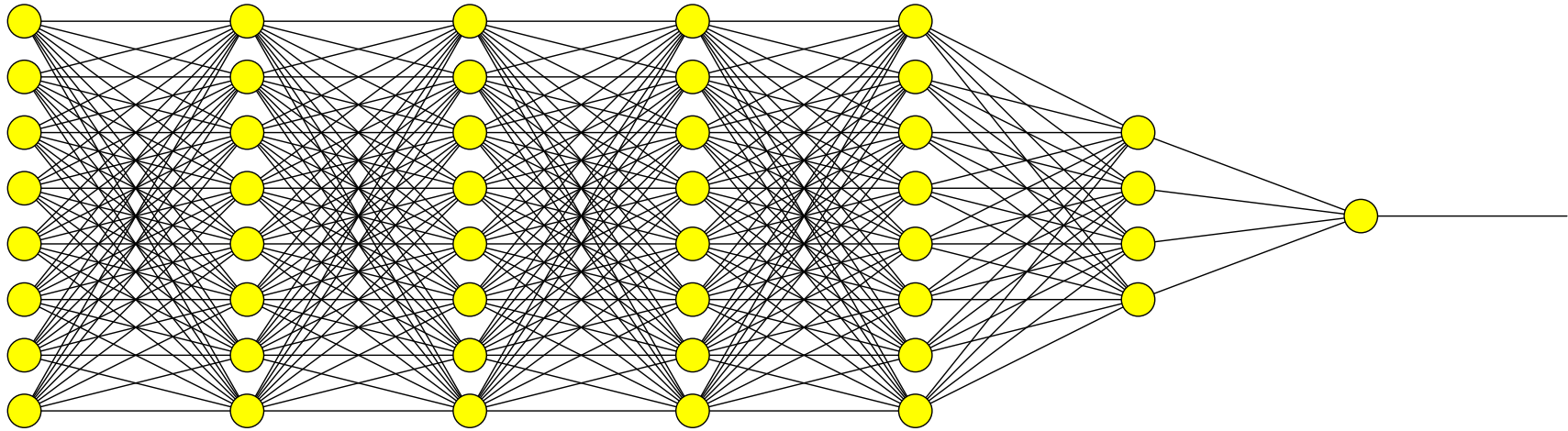
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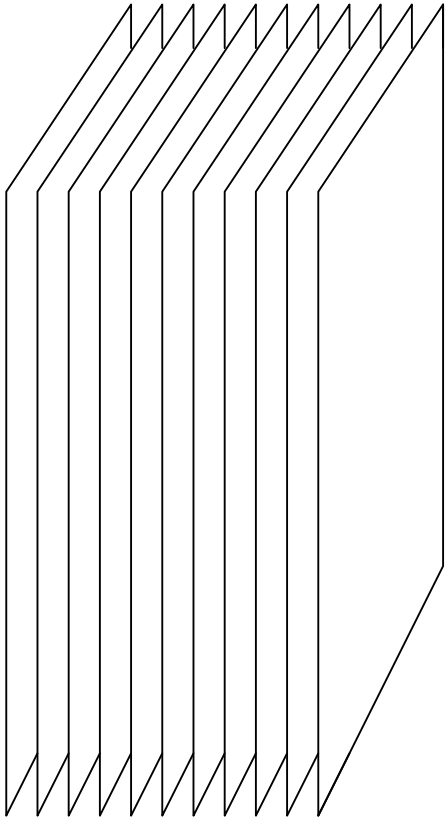


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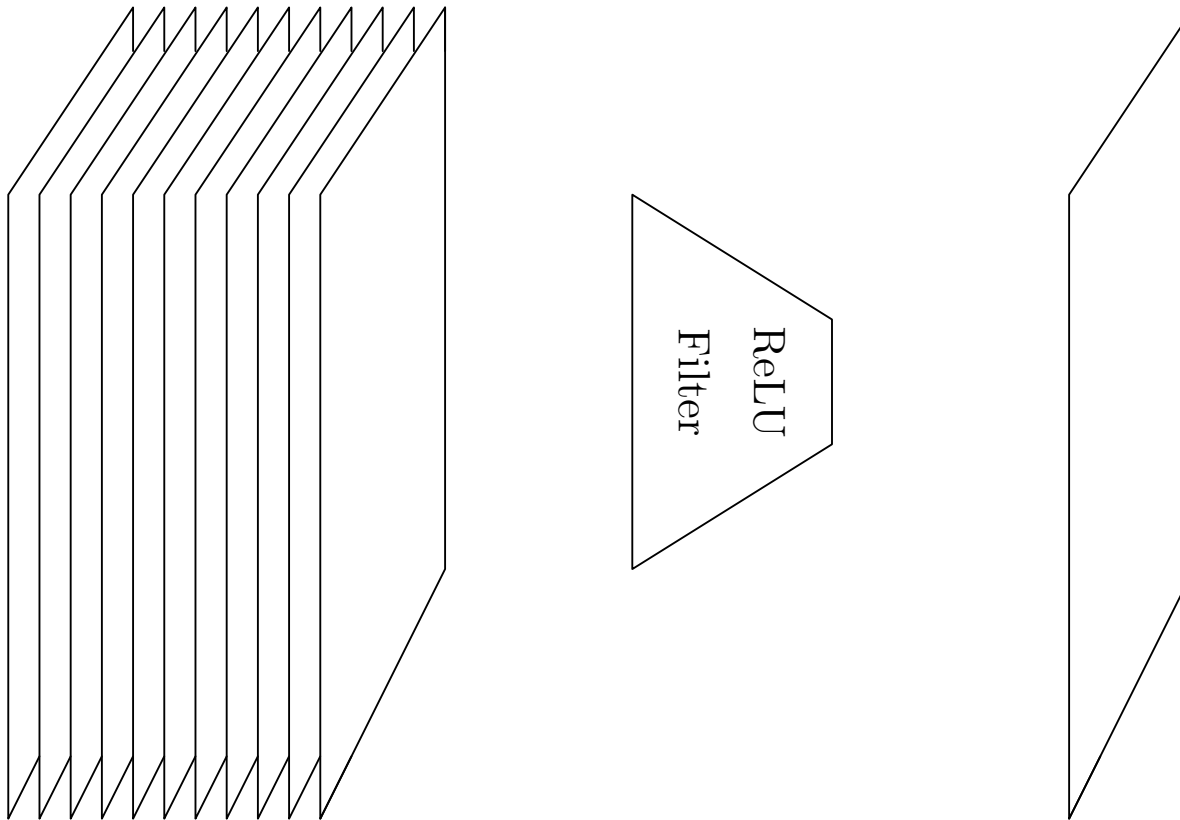


Acts as a strong form of regularisation

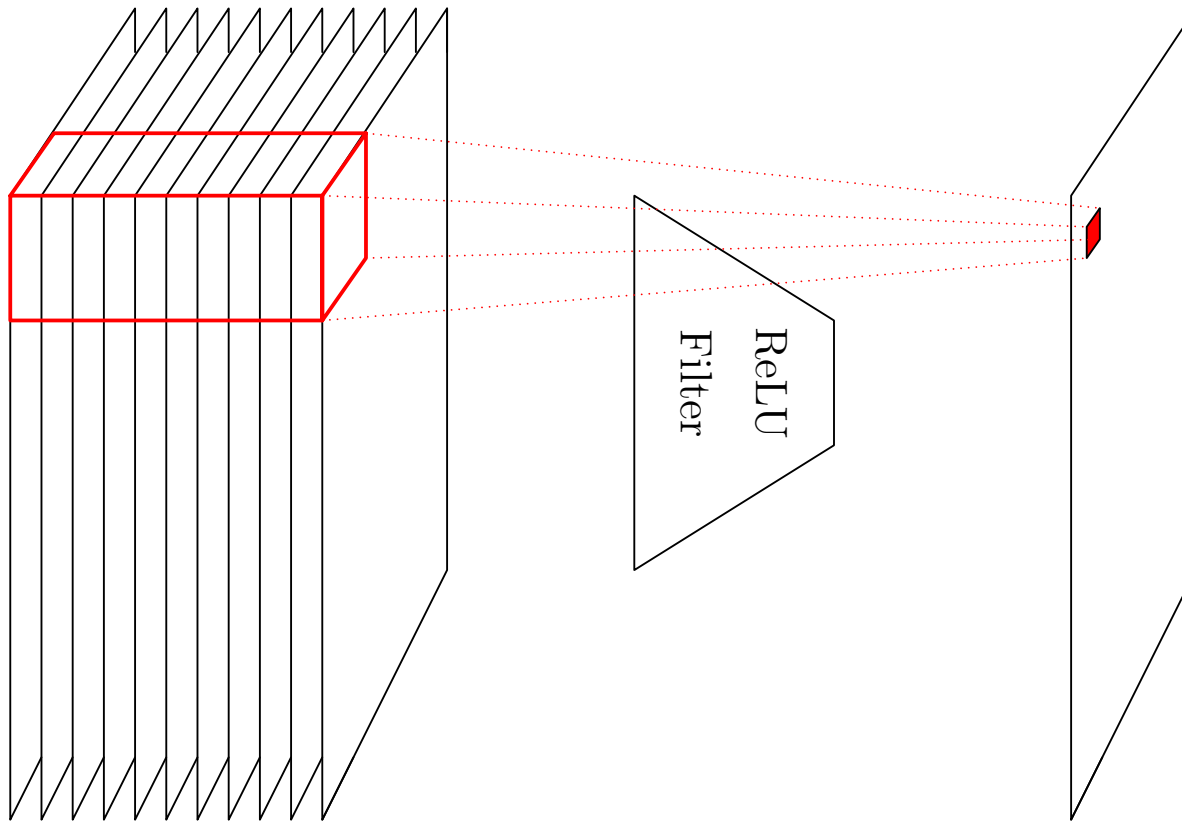
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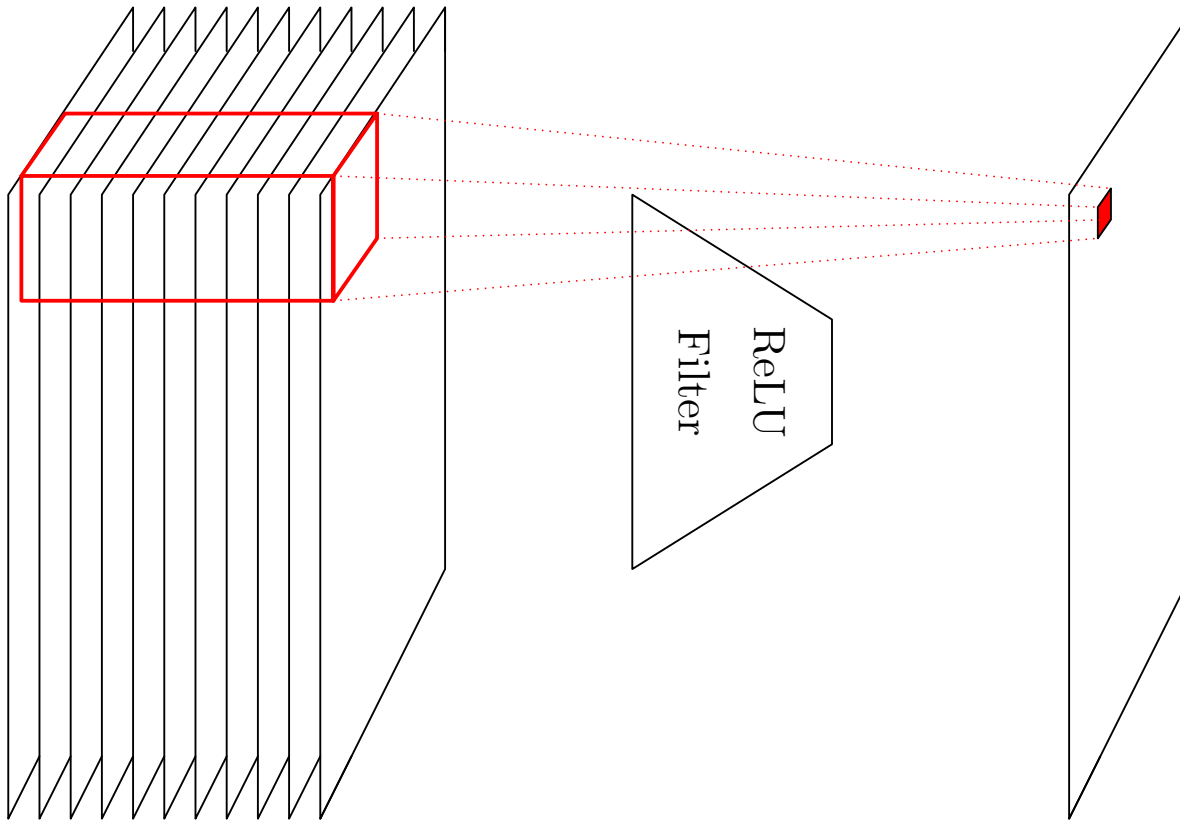
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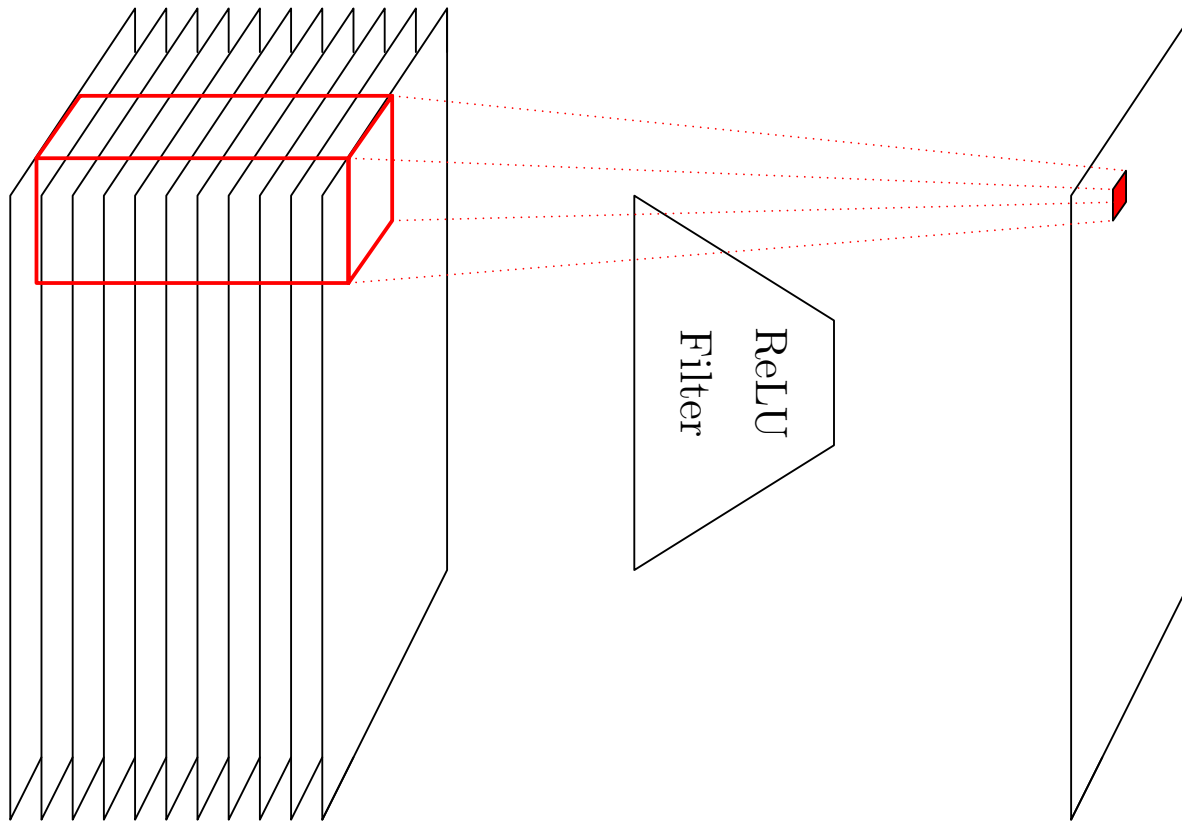
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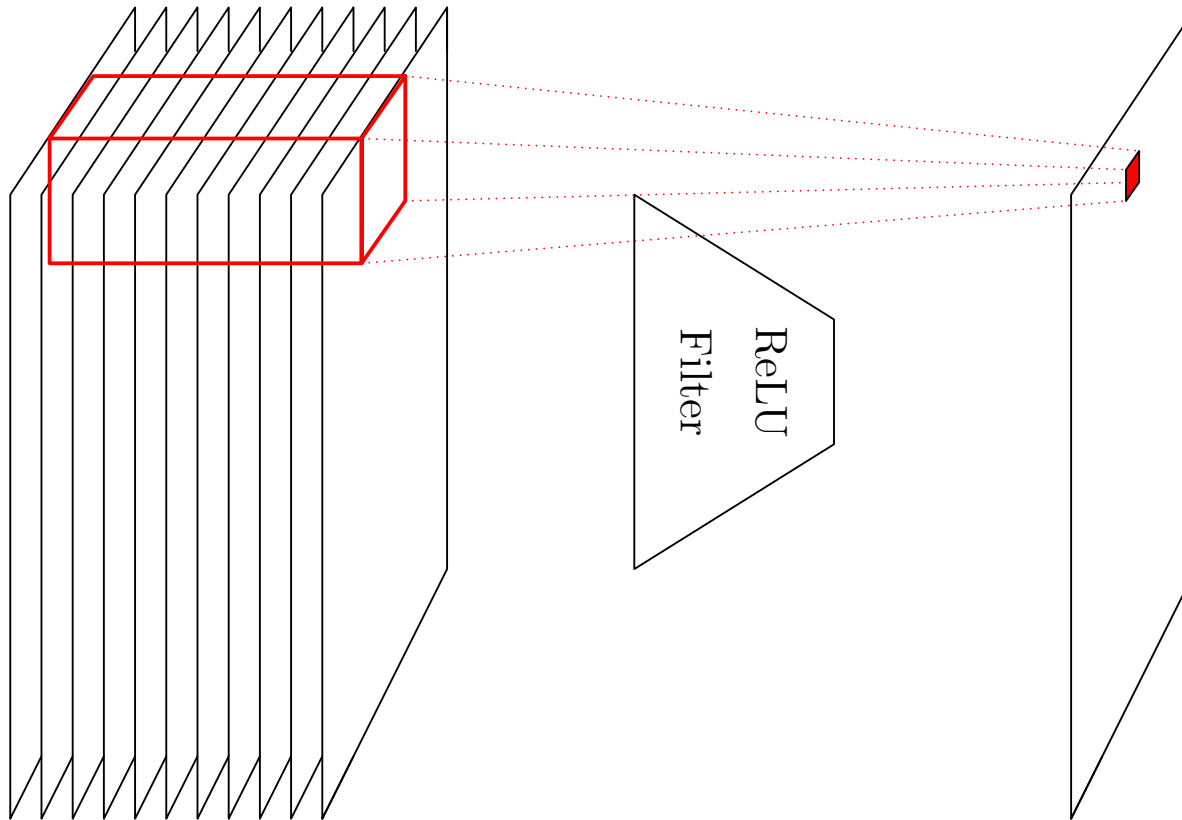
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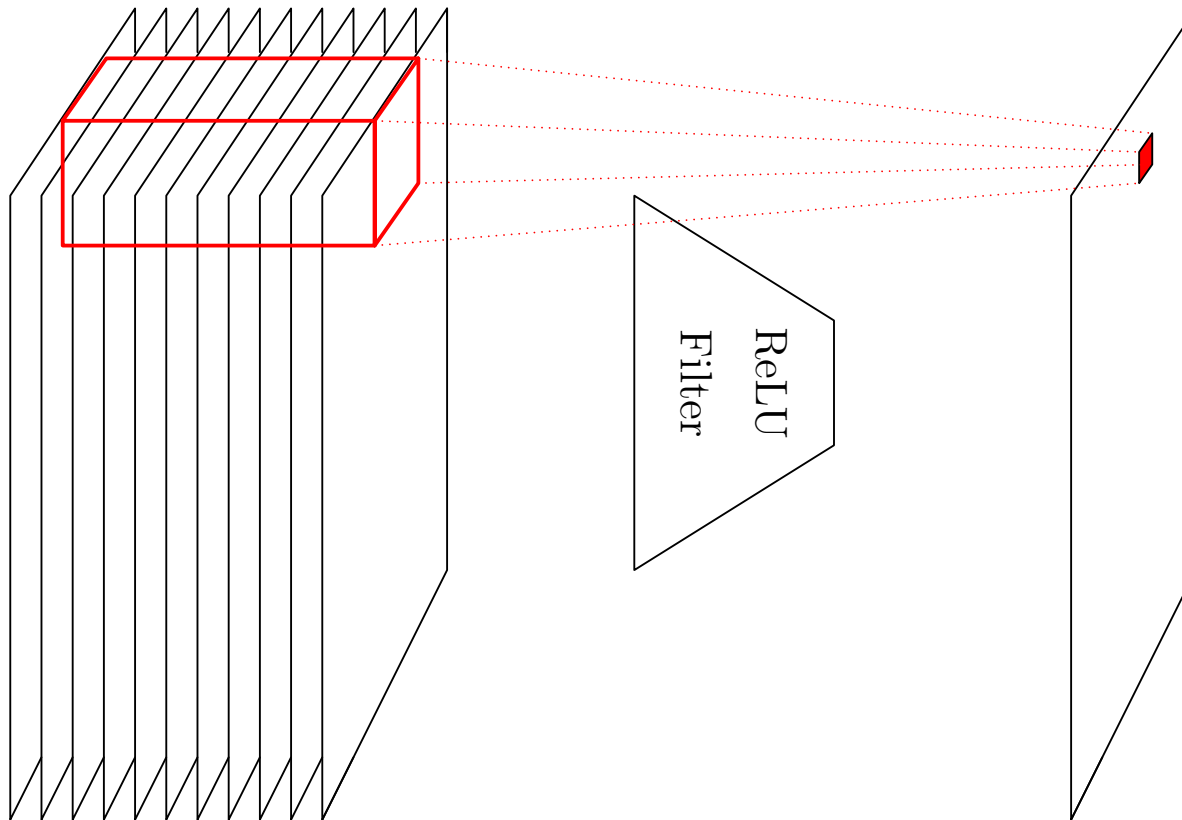
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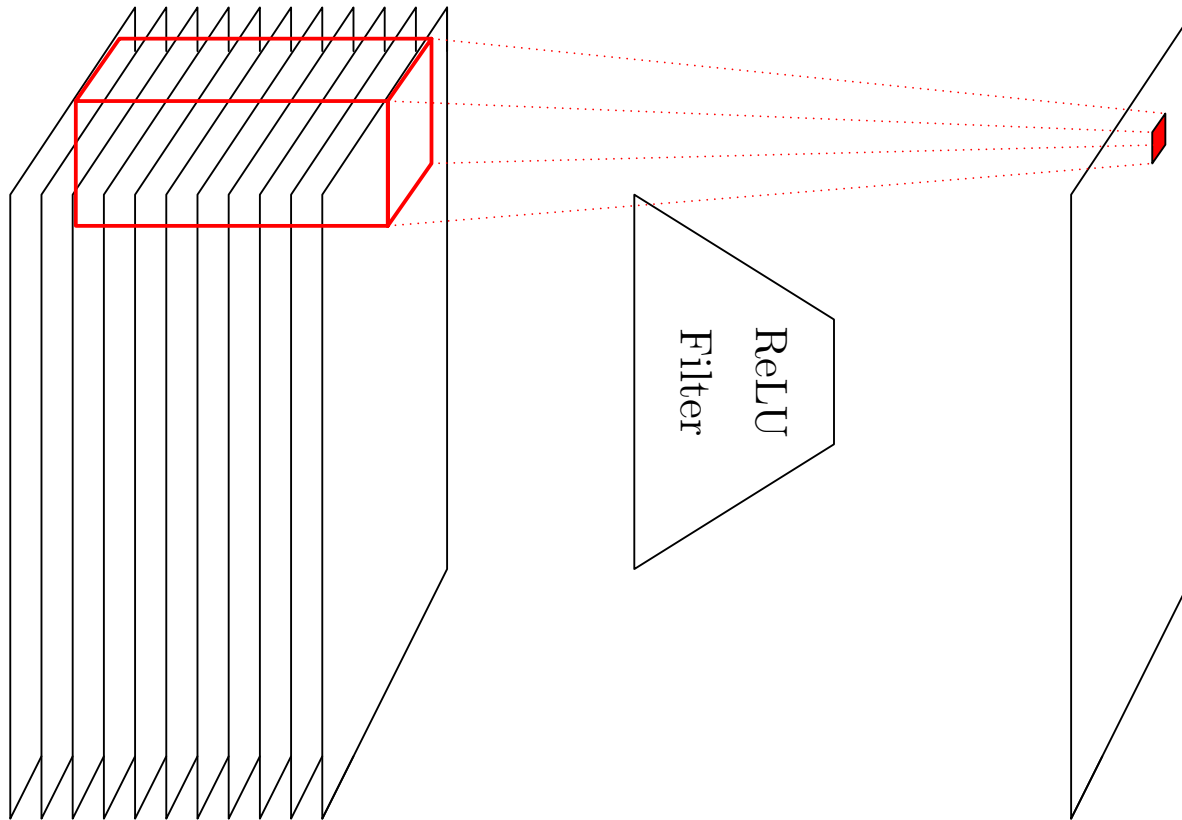
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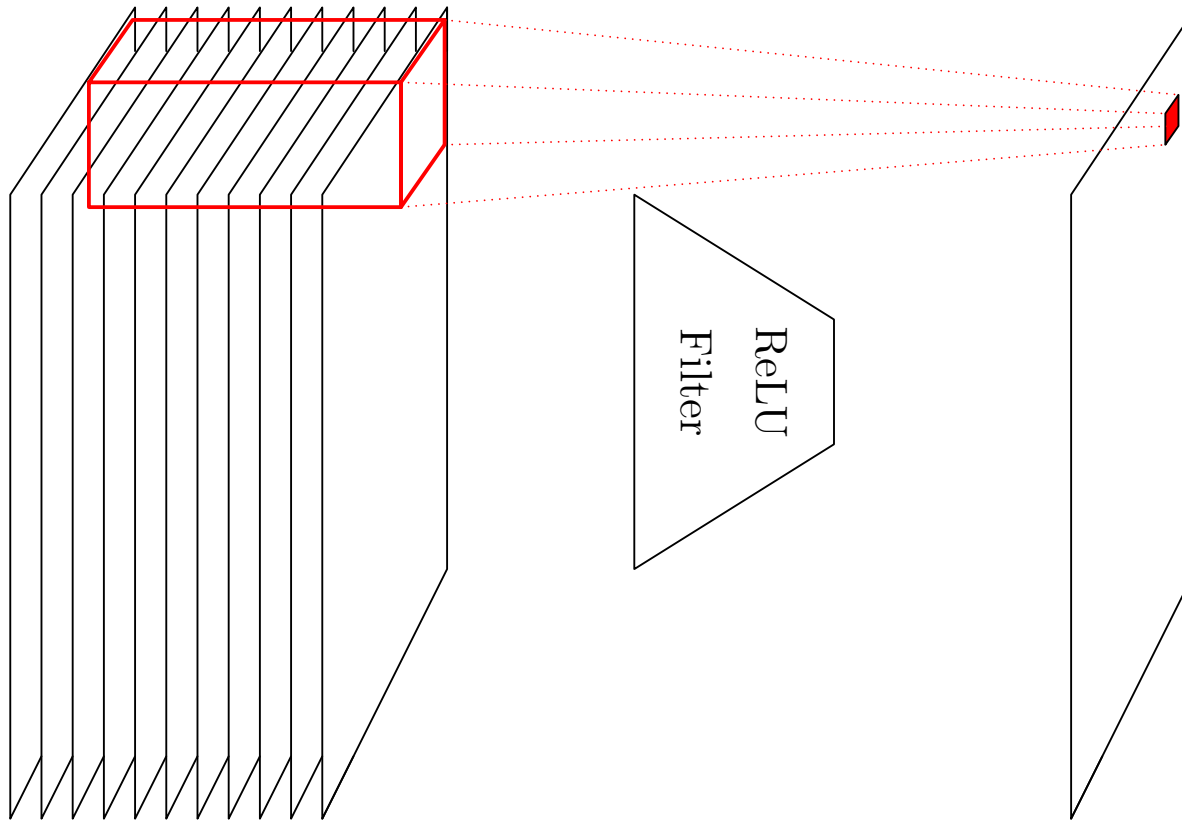
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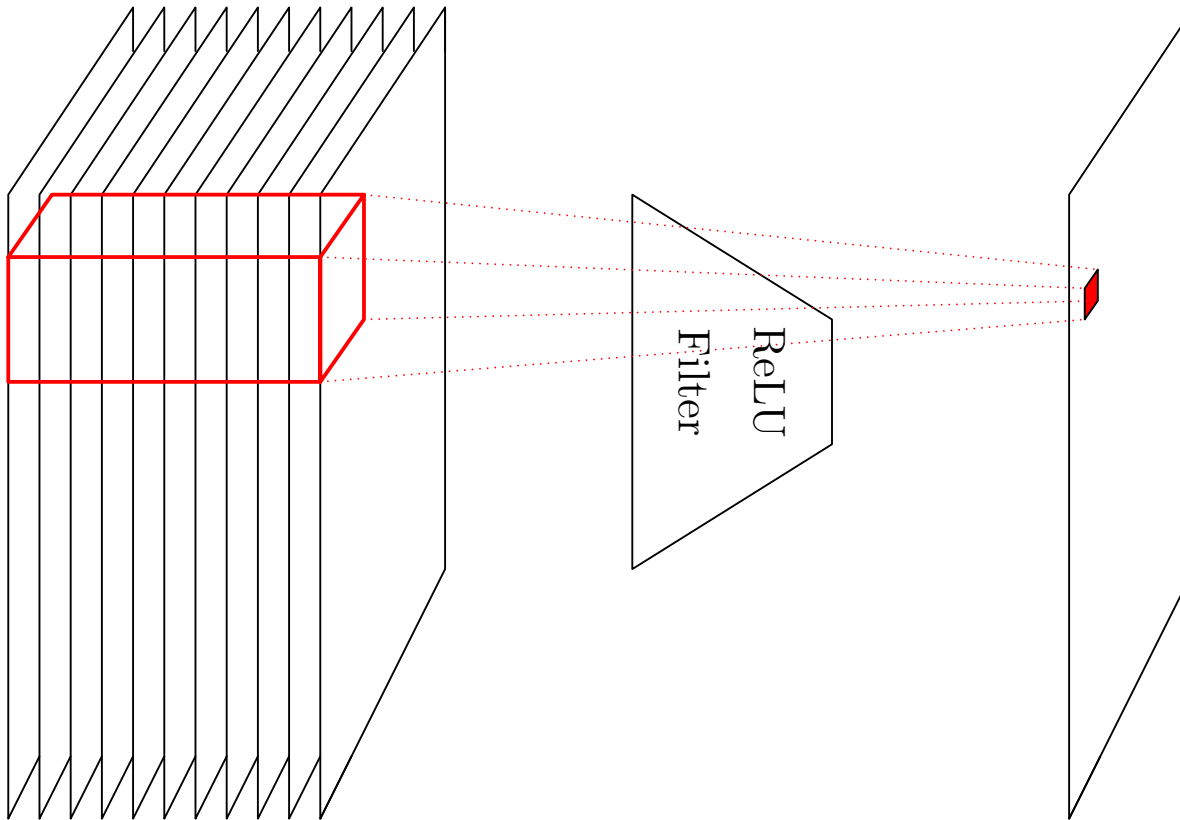
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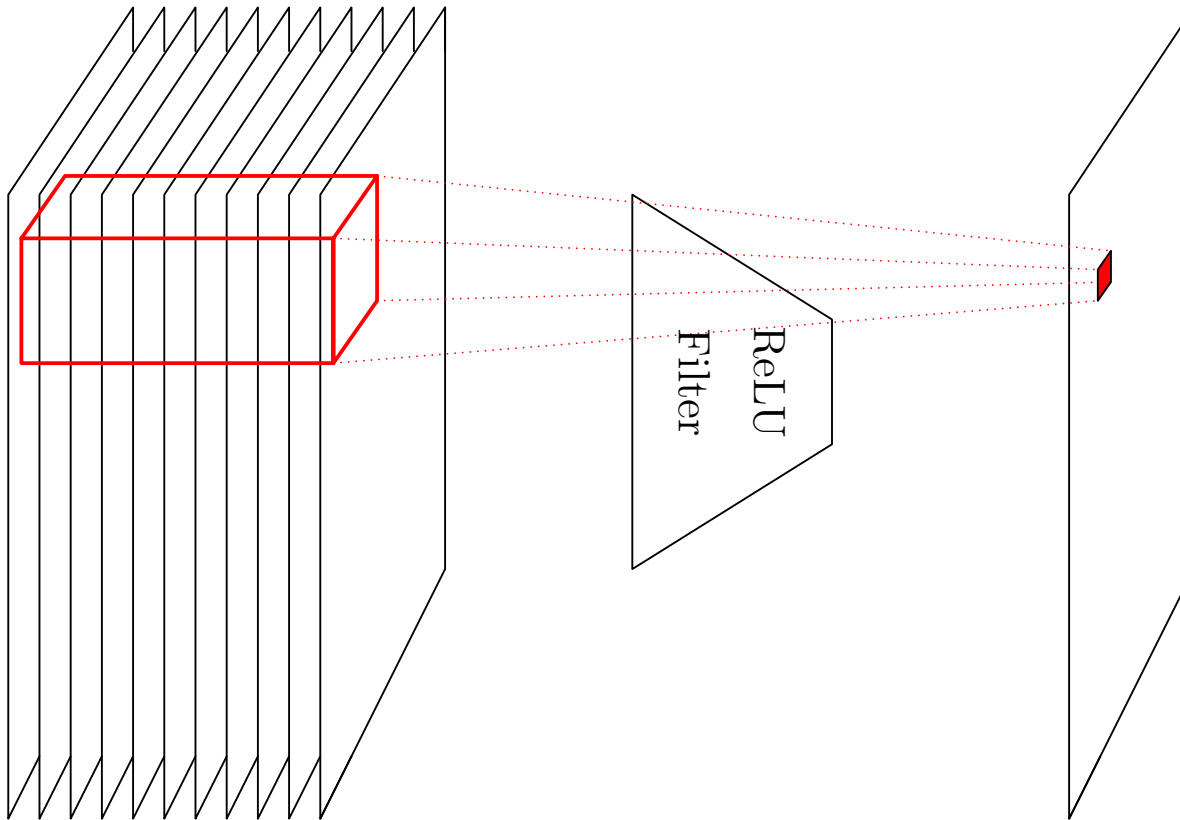
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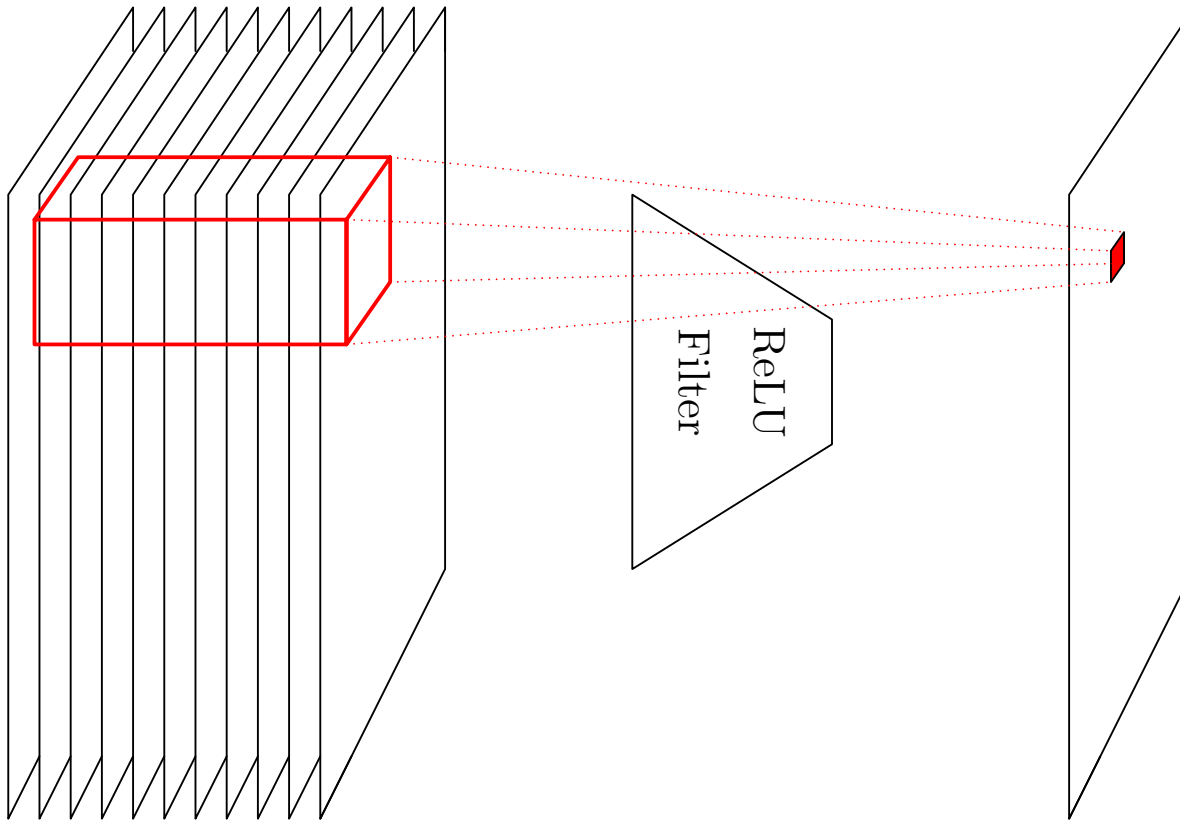
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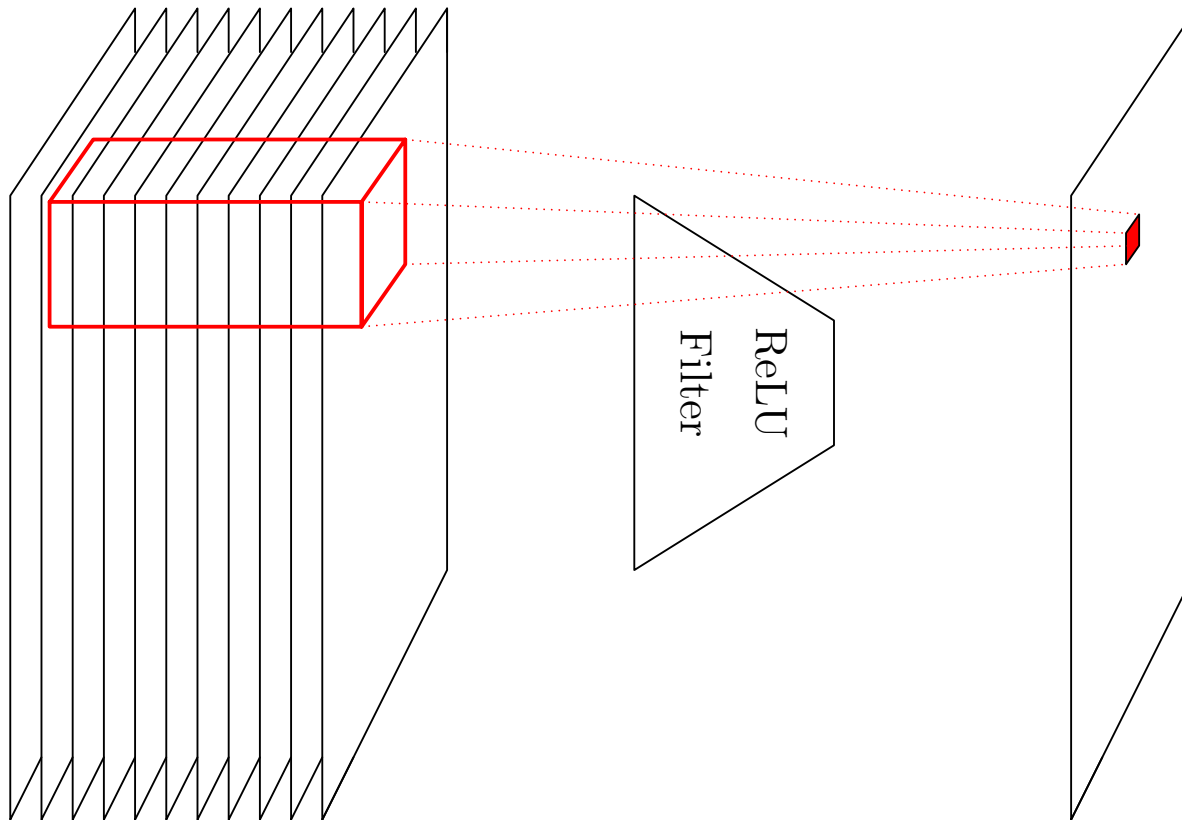
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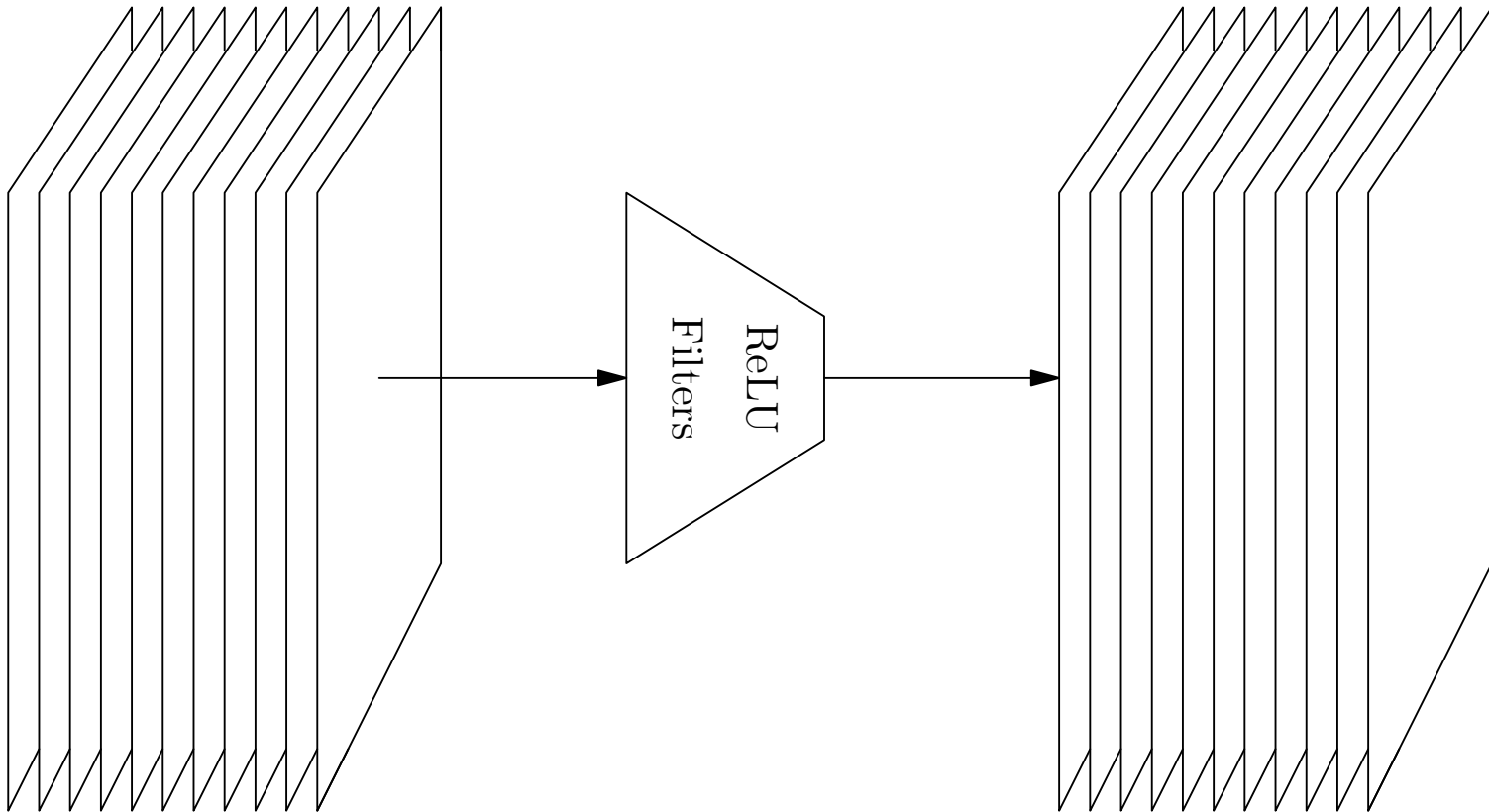
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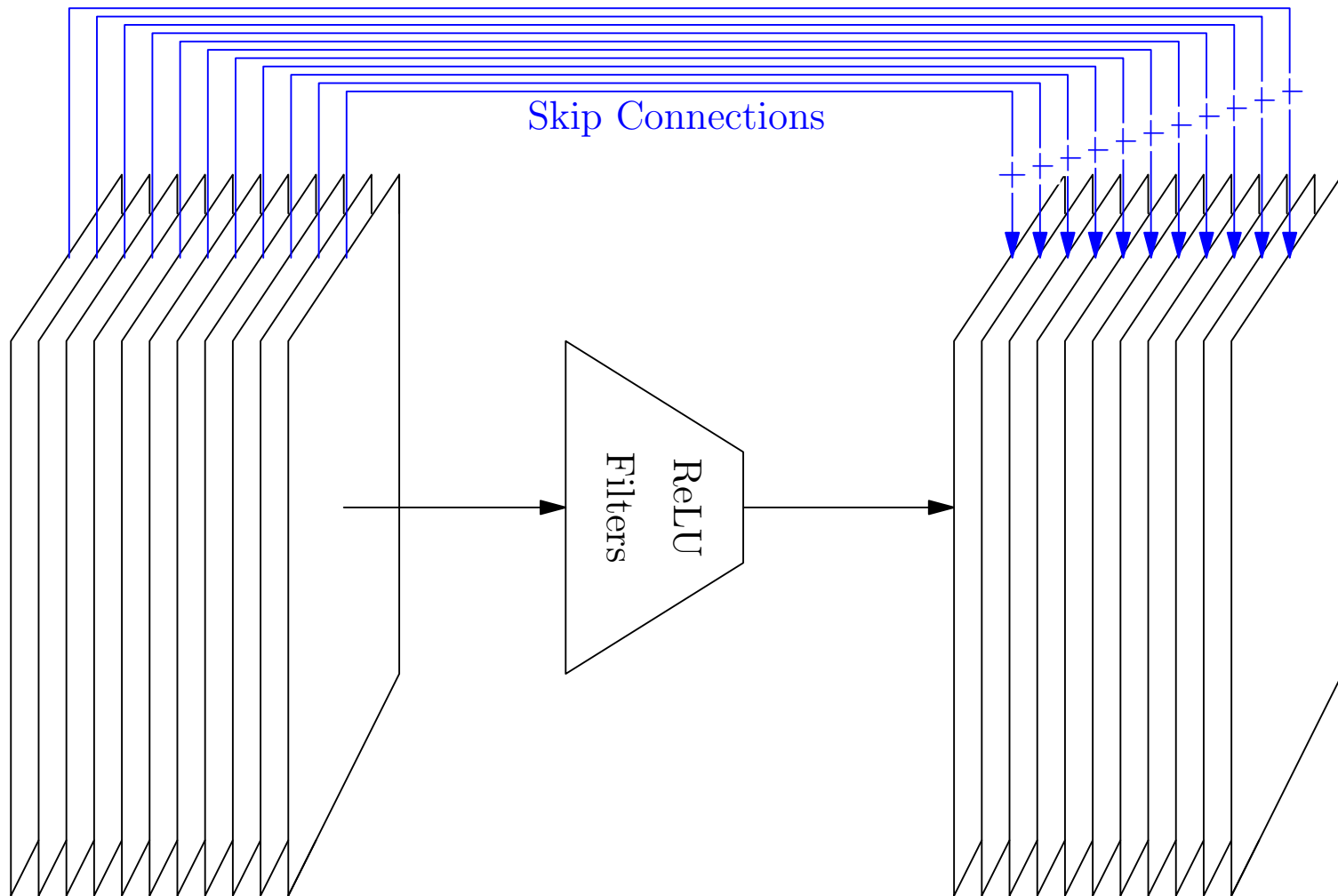
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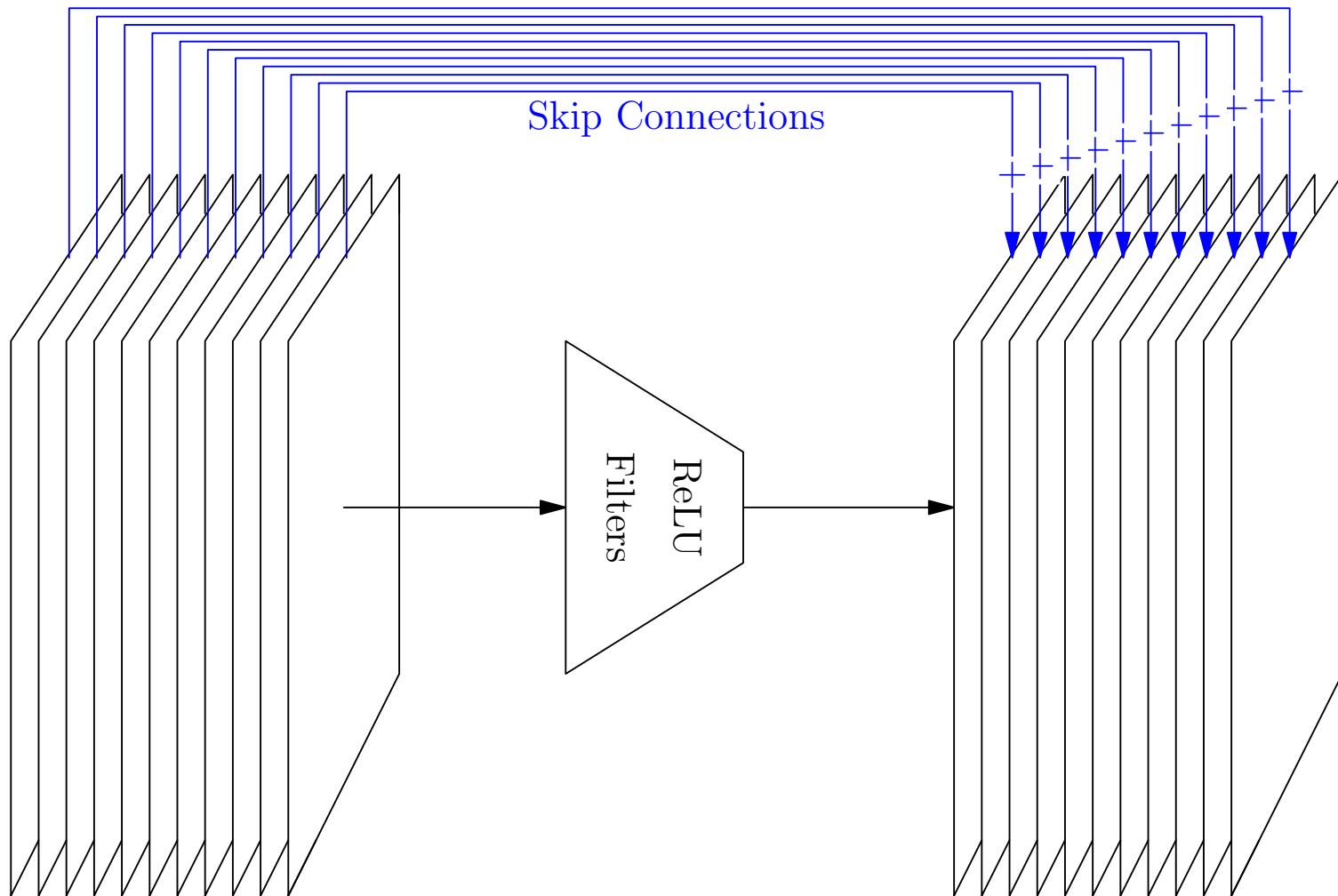
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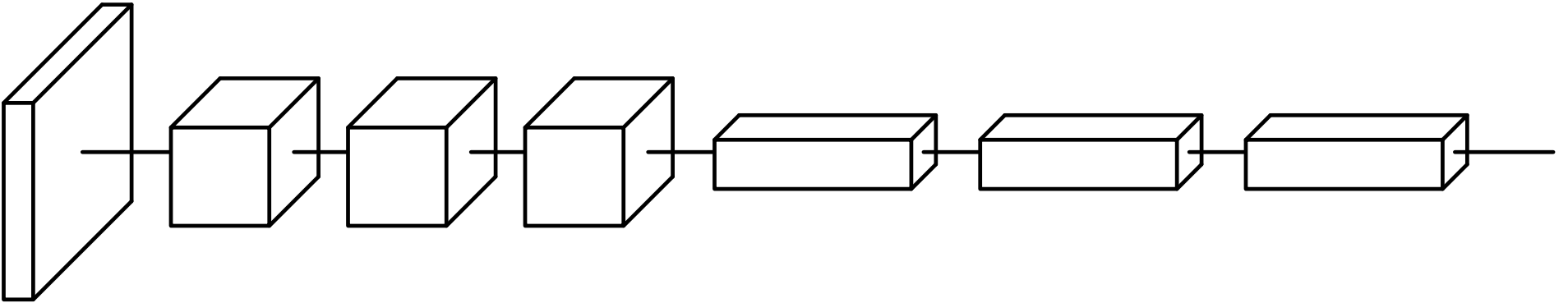


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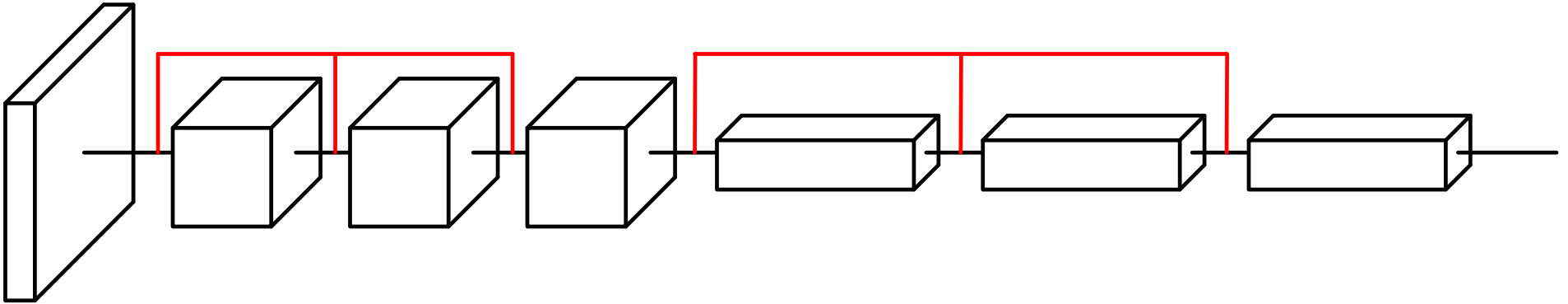


Breaks symmetry, improves signal and gradient vanishing

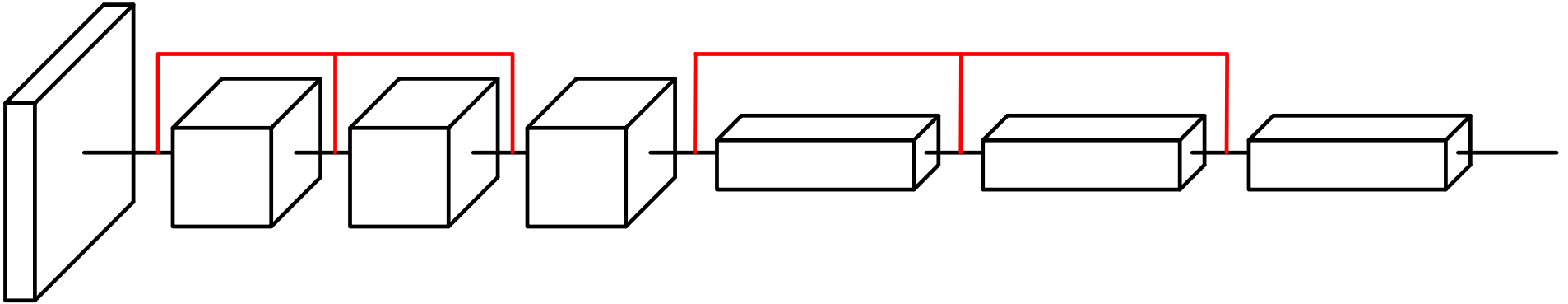
New Architectures: ResNet



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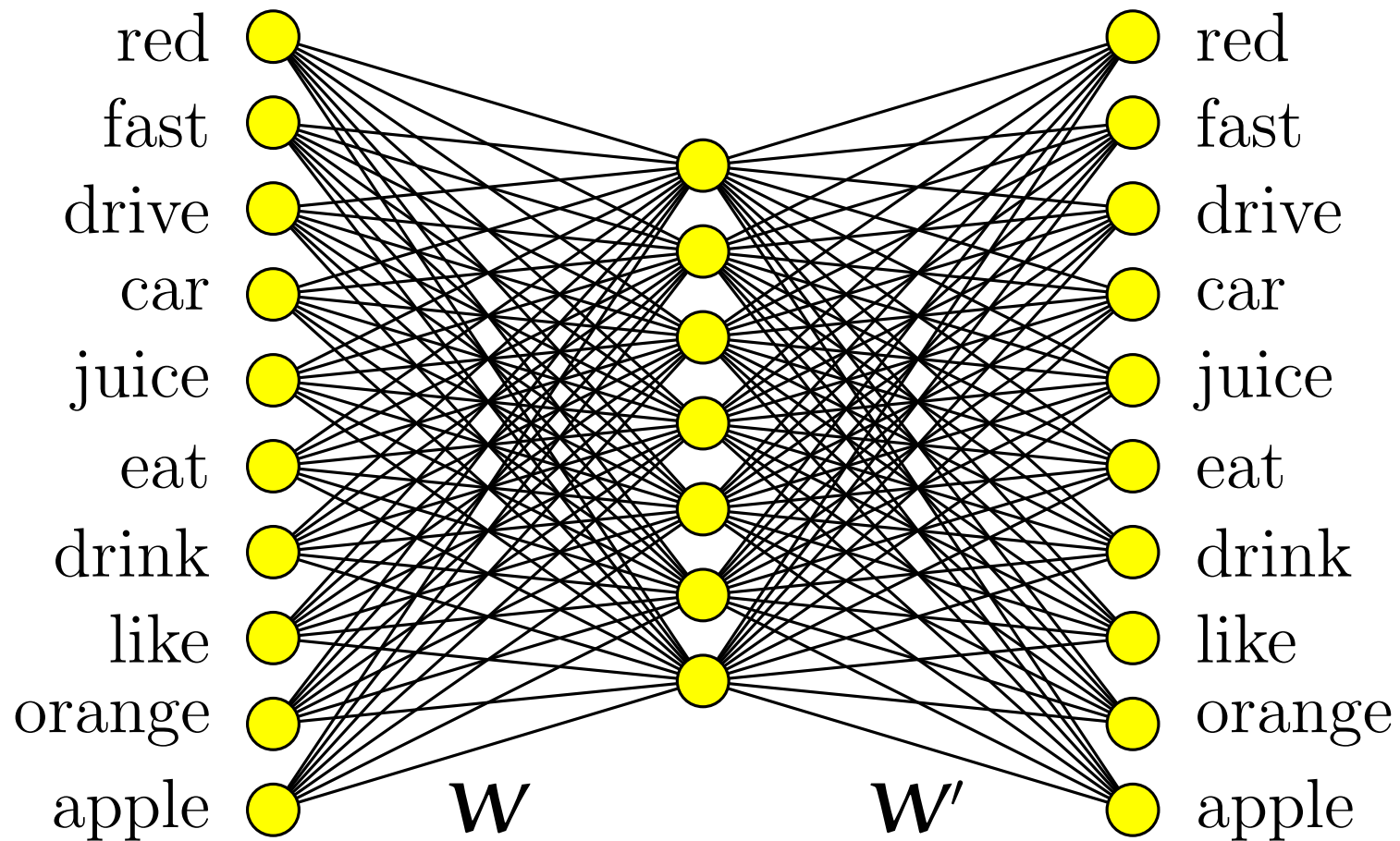


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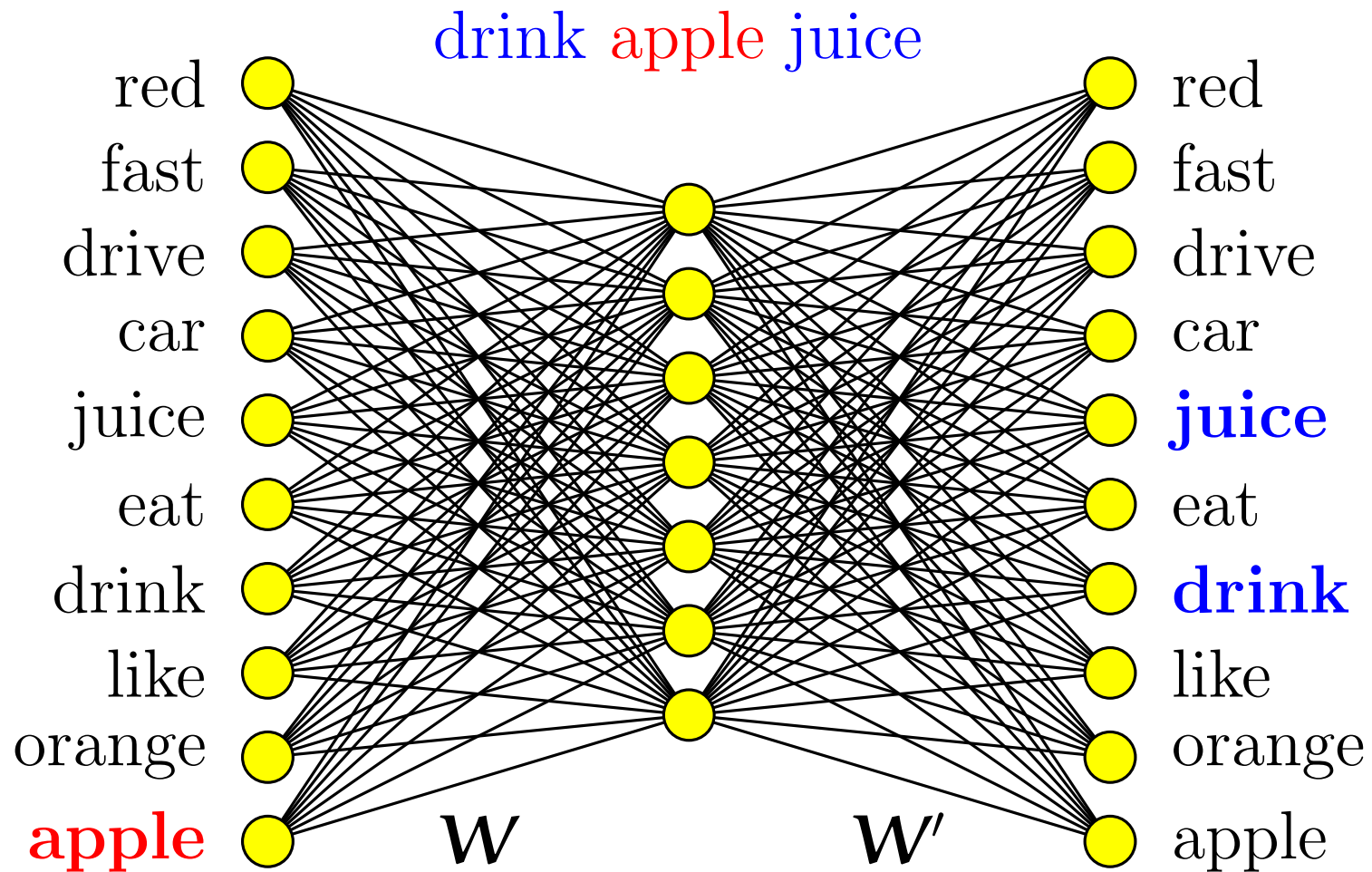


Used with batch normalisation, allowing very deep networks

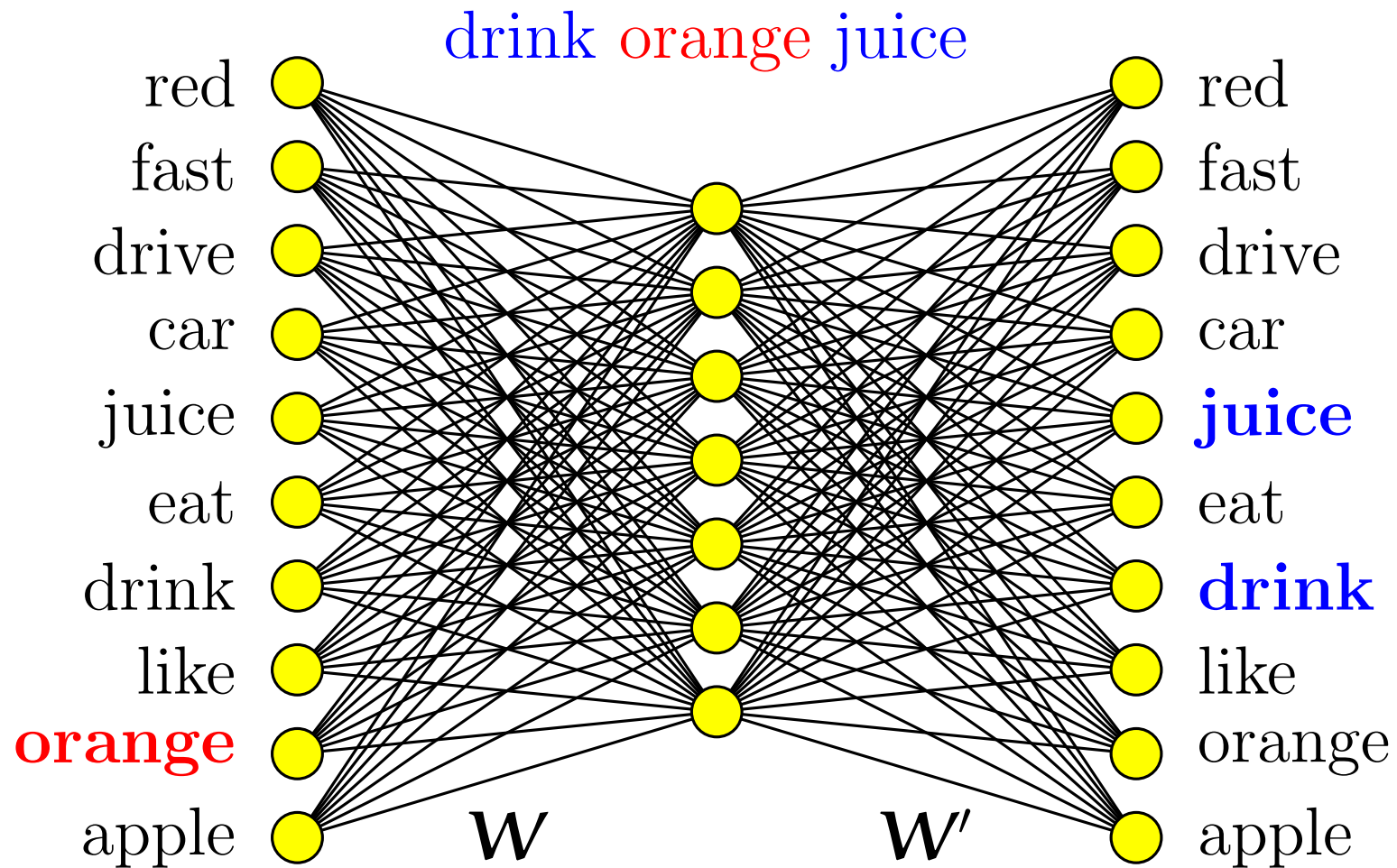
New Ideas: Embeddings



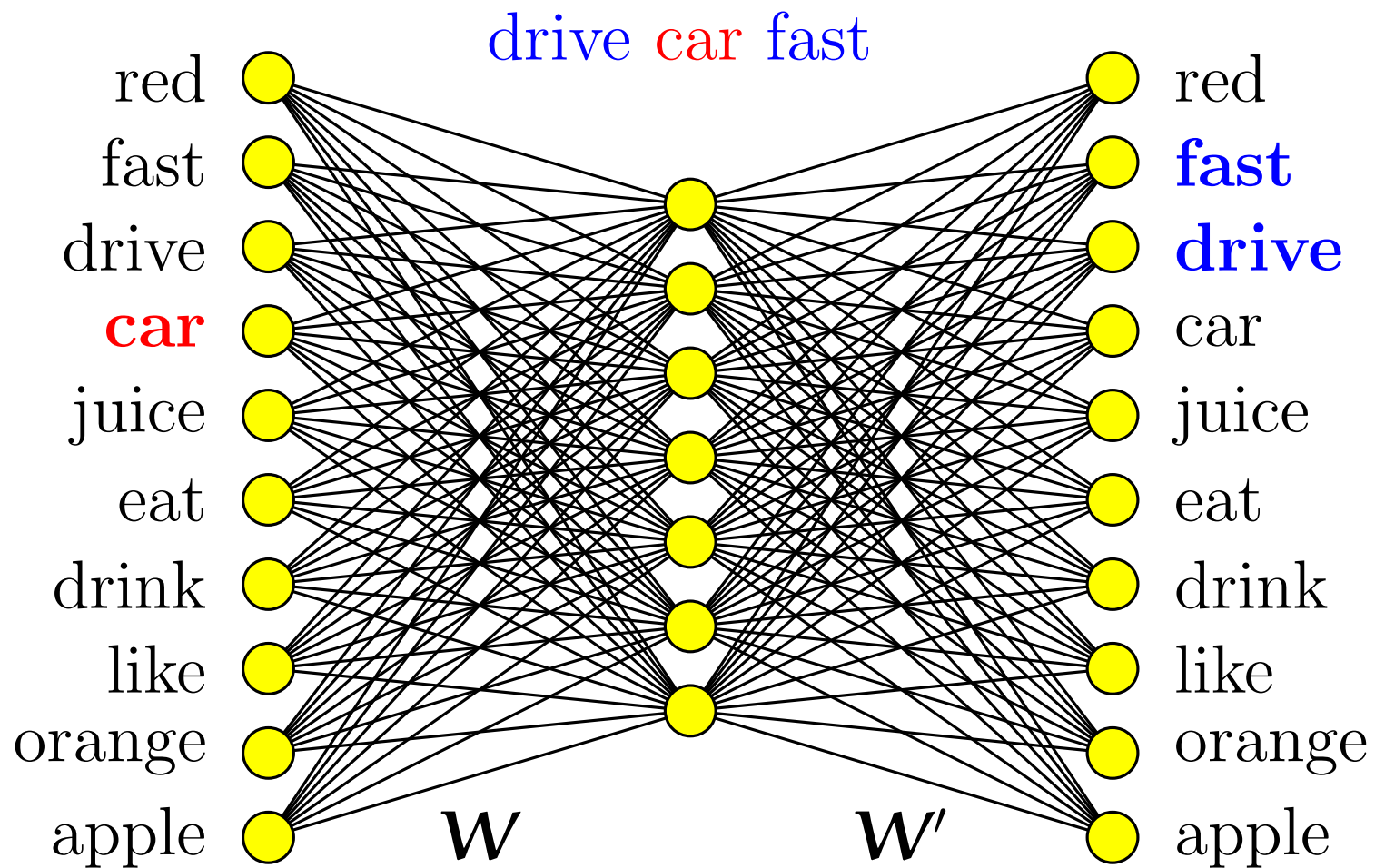
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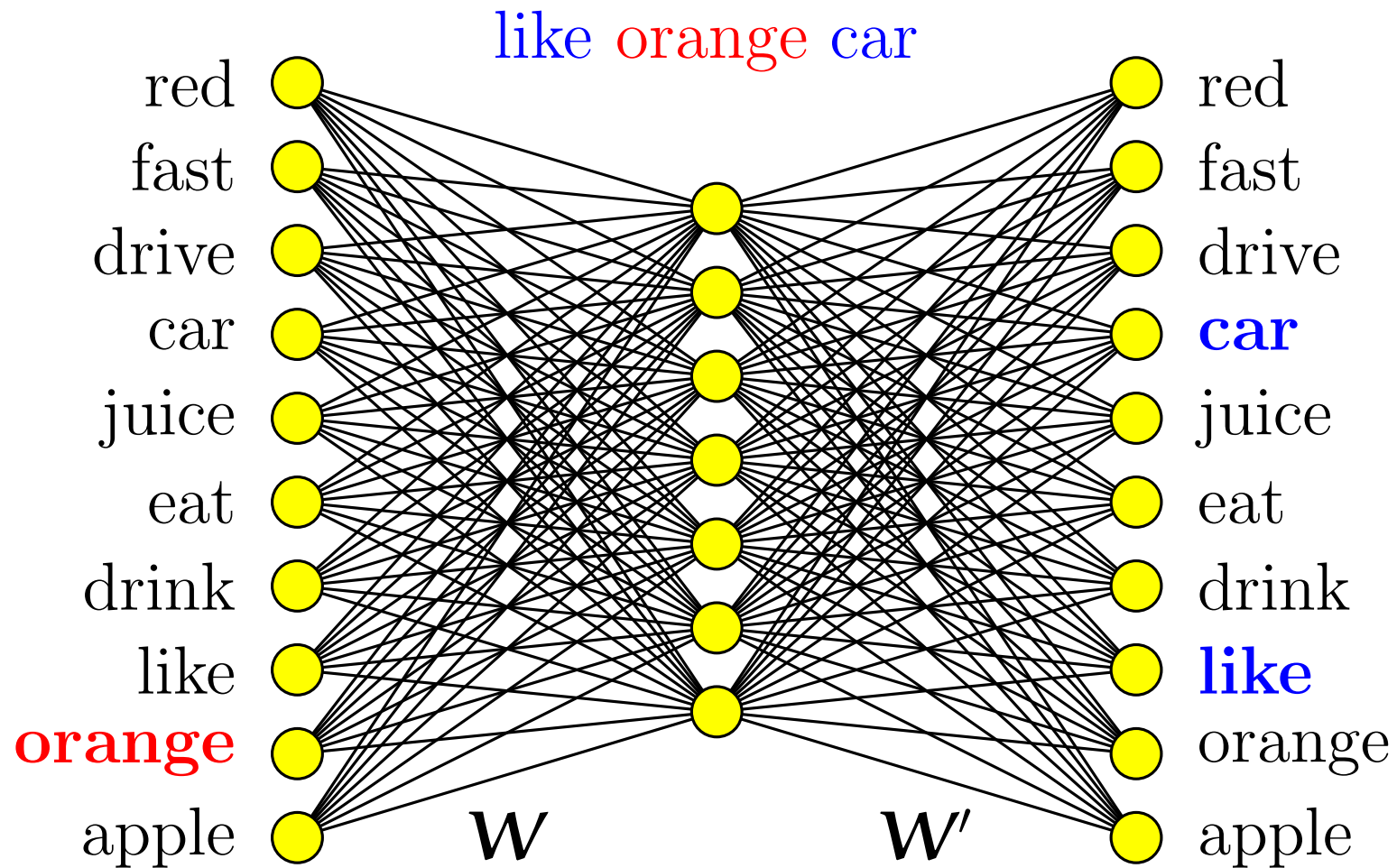
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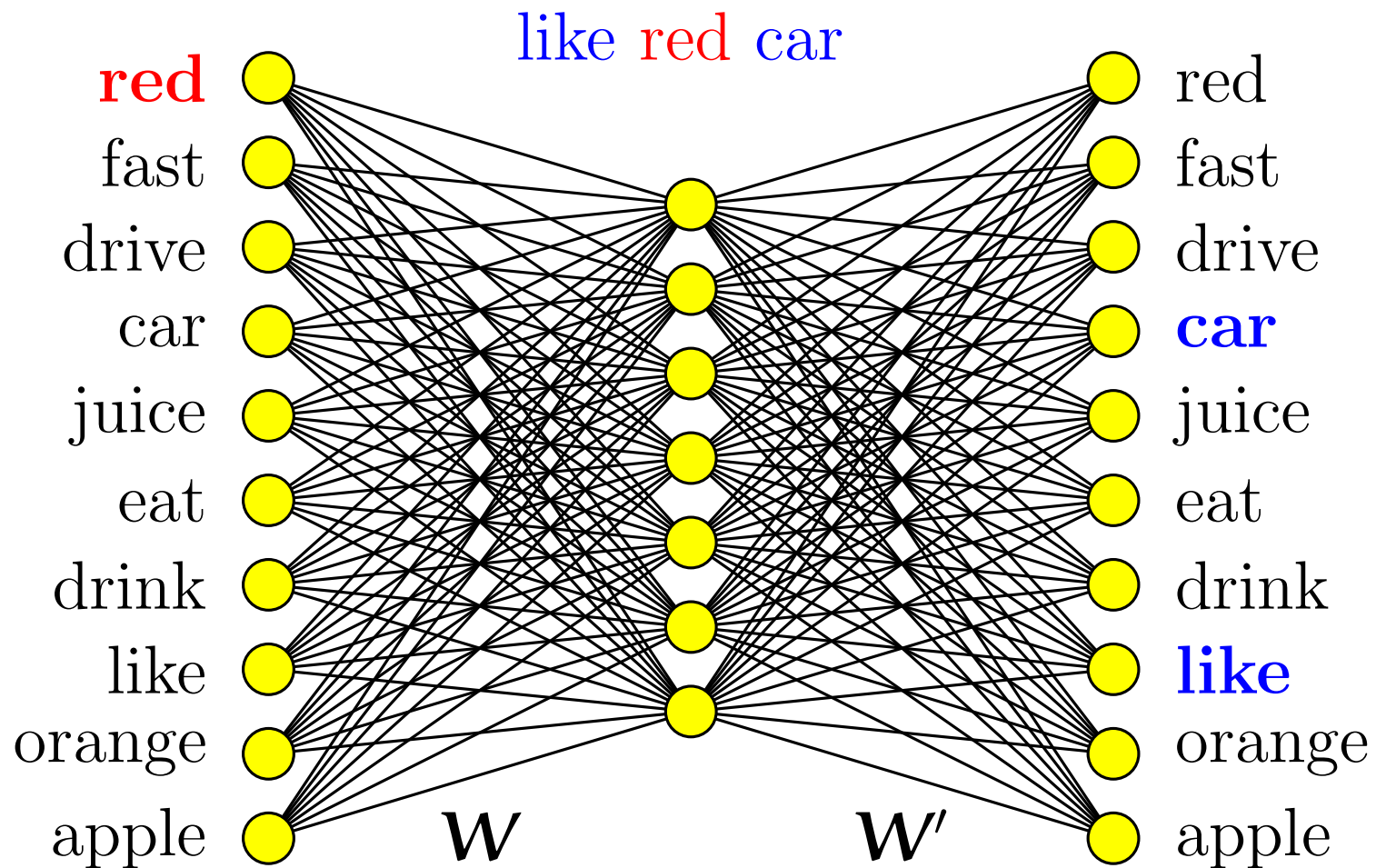
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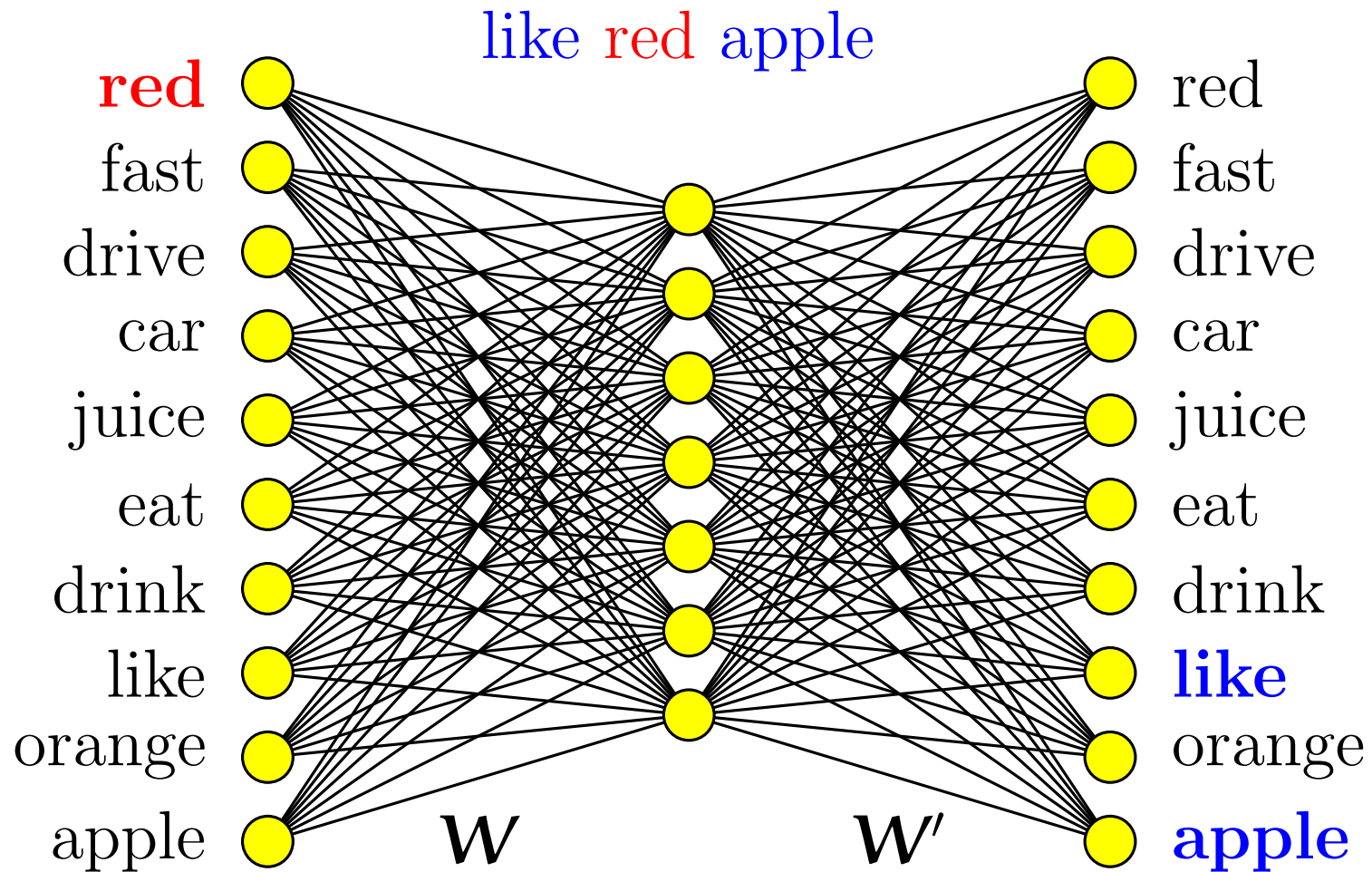
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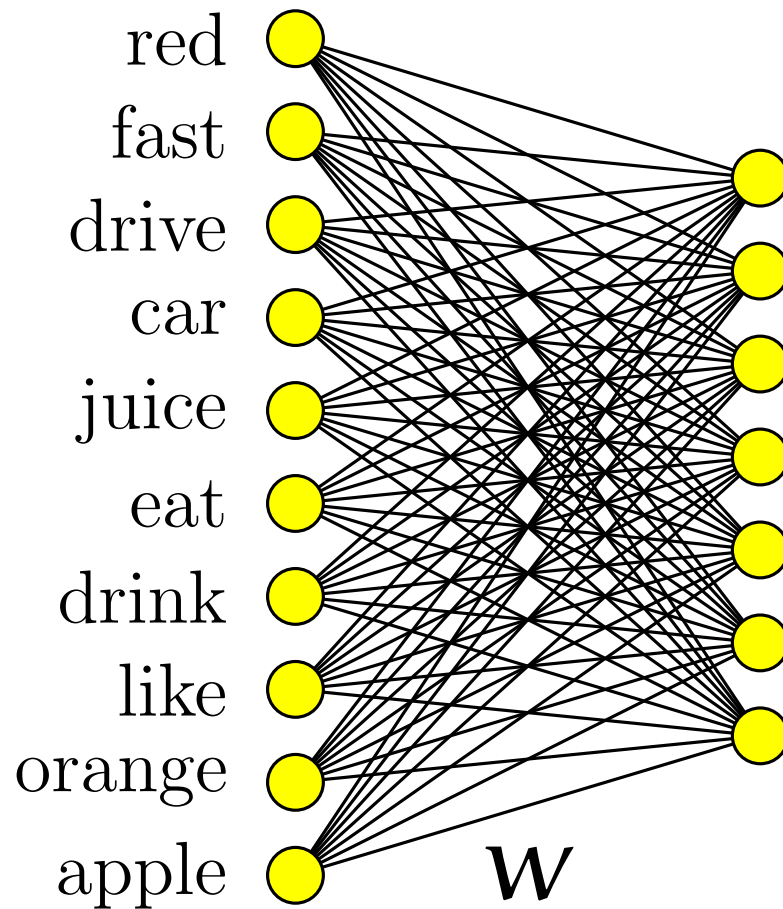
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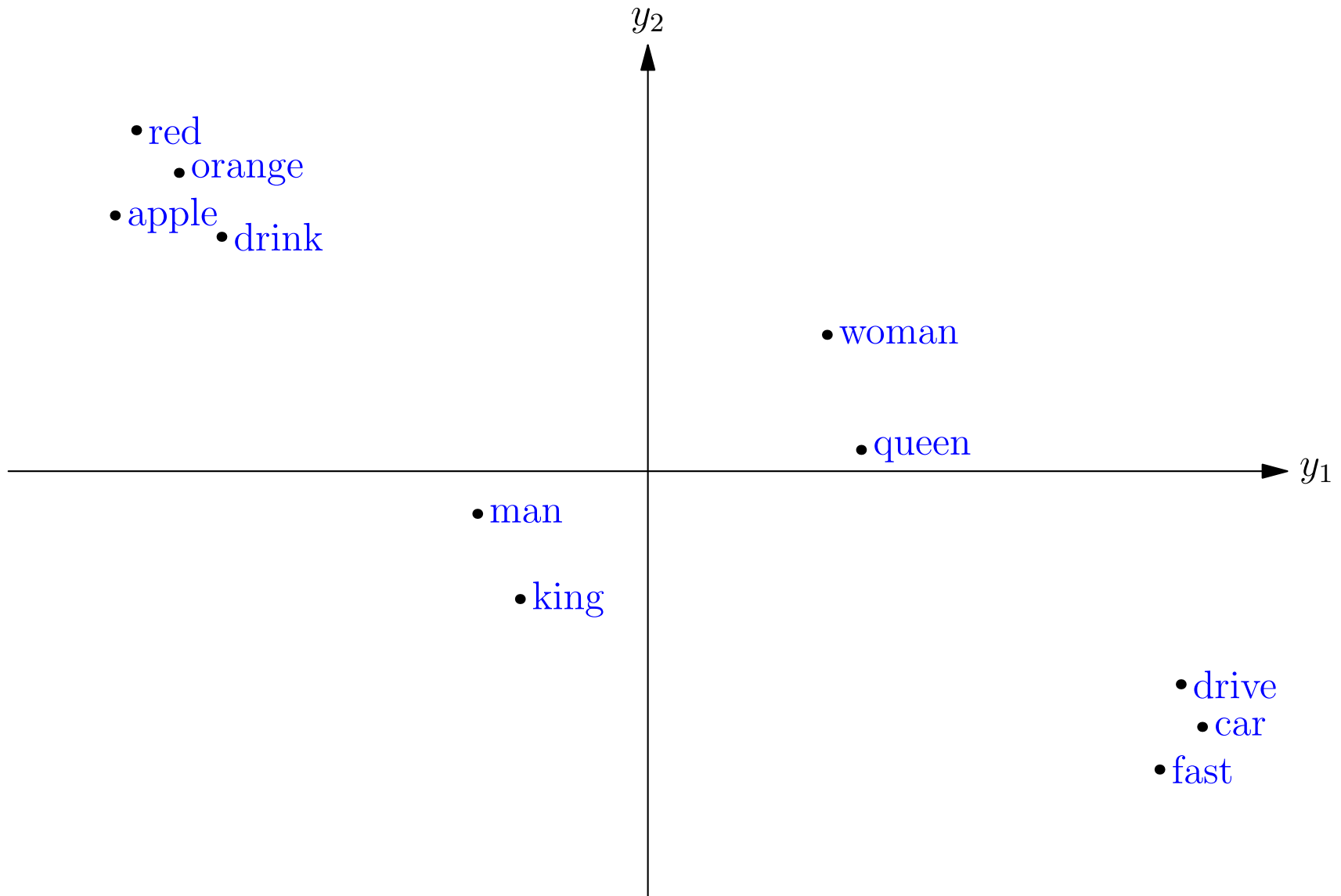
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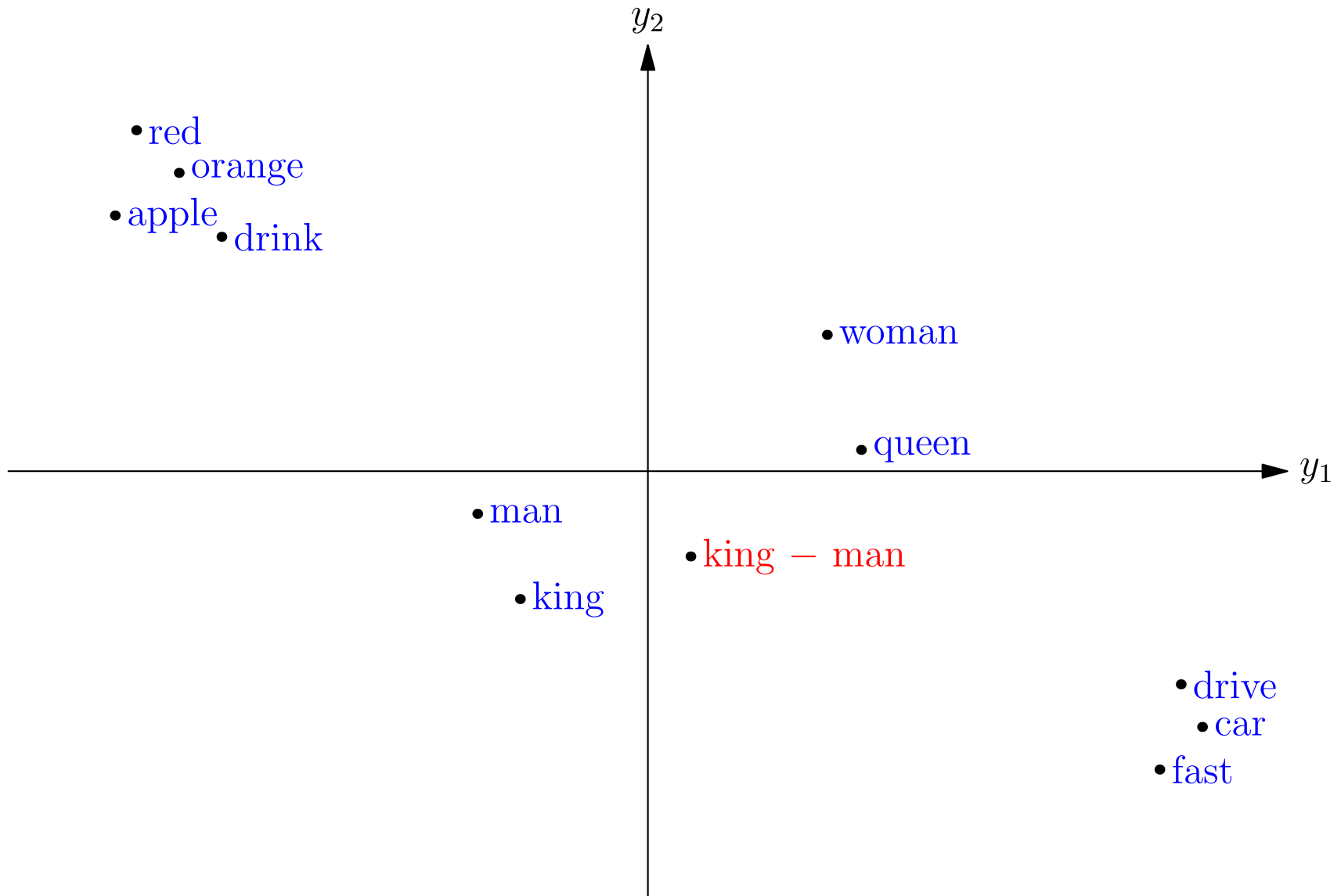
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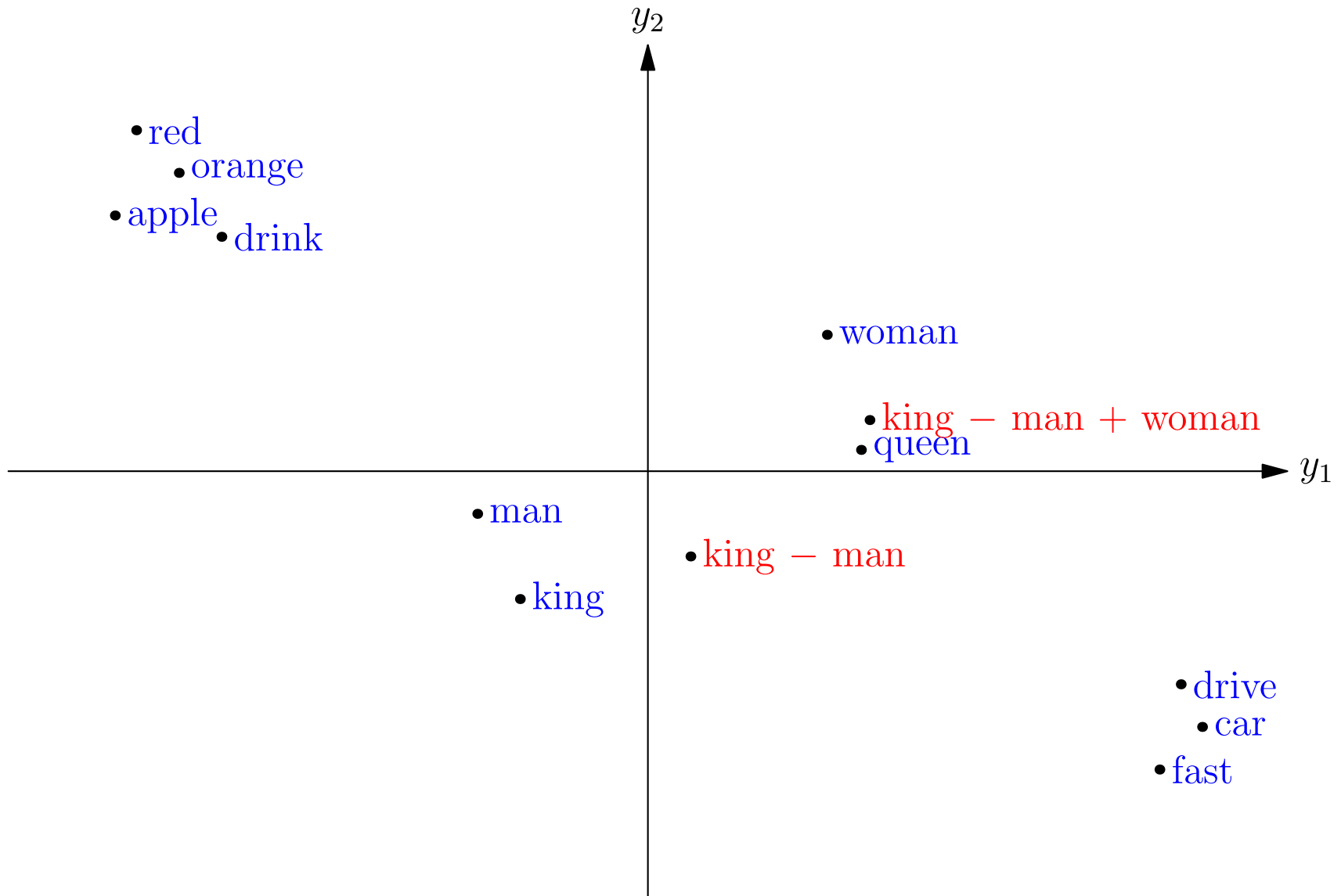
Word Embedding Space



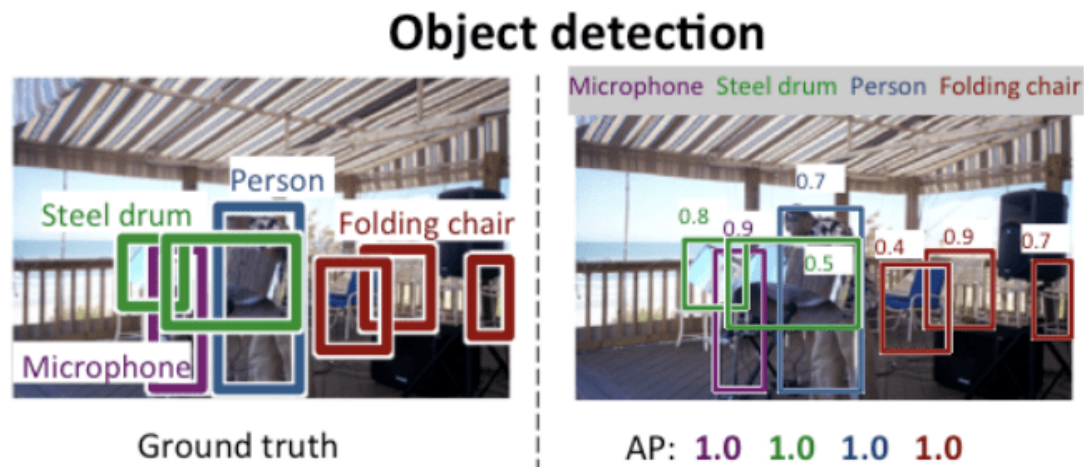
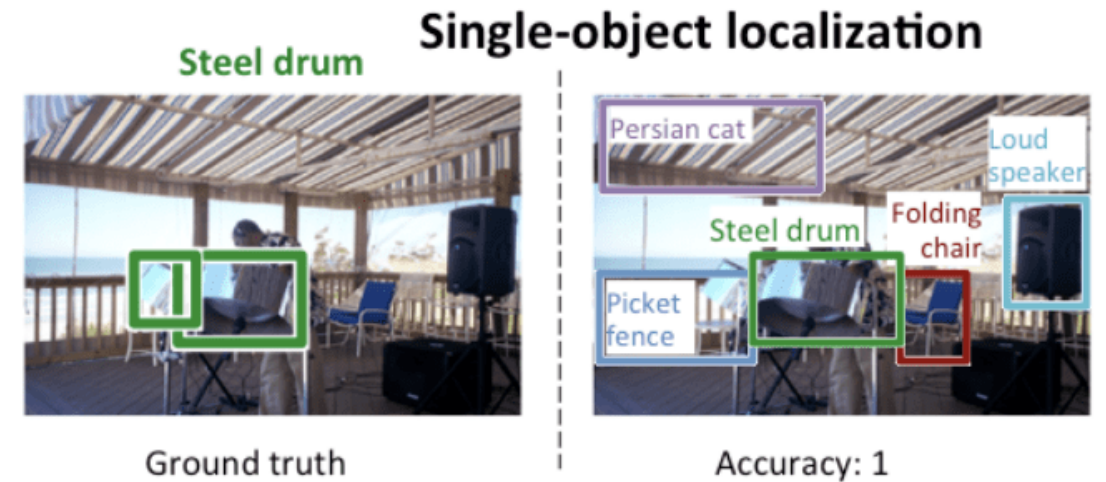
Word Embedding Space



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New Applications: Object Localisation



Fast-RCNN, Faster-RCNN, YOLO, Mask-RCNN

New Applications: Image Segmentation



U-Nets

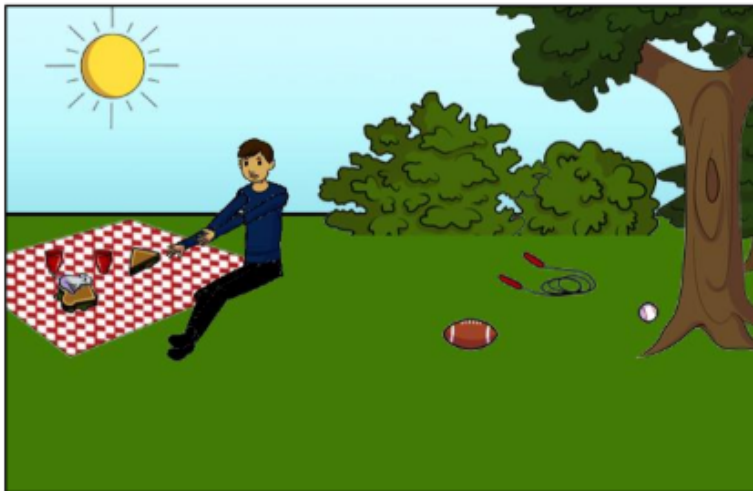
New Applications: VQA



What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?

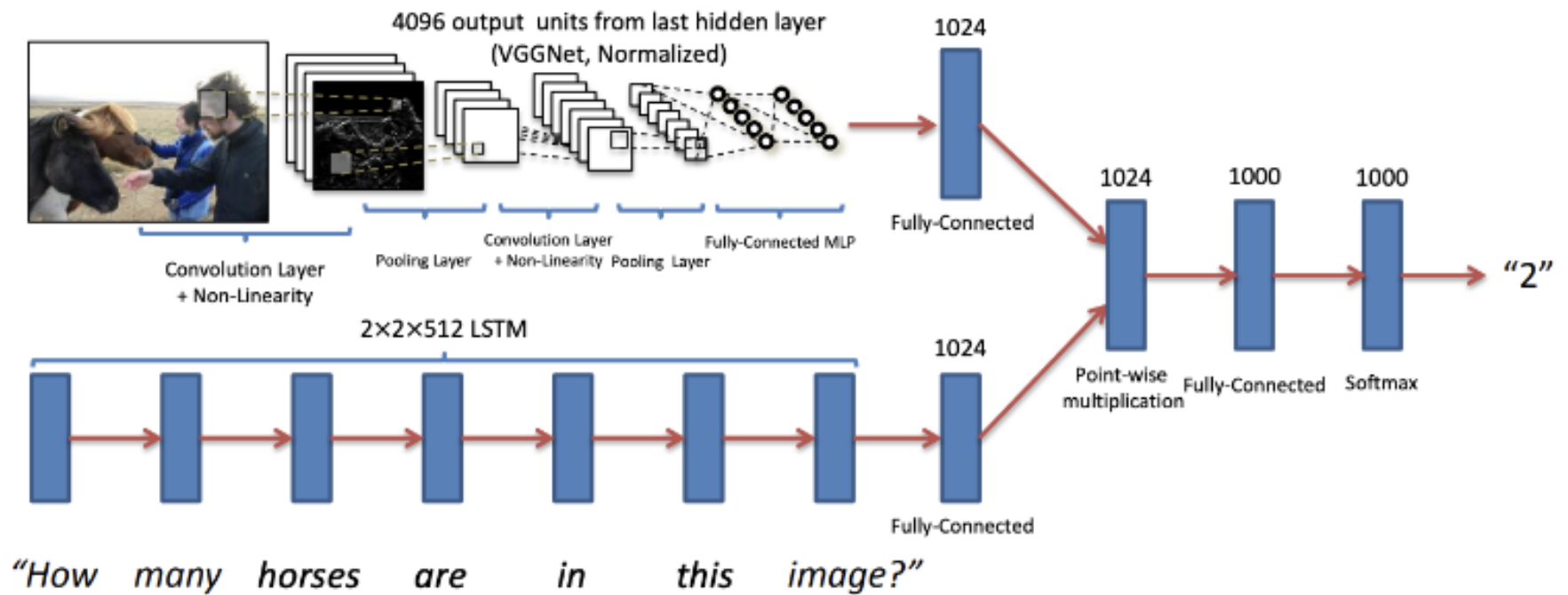


Is this person expecting company?
What is just under the tree?



Does it appear to be rainy?
Does this person have 20/20 vision?

Combining Networks



Towards Generative AI

- One of the sticking points was the need for labelled data sets
- To avoid this the community turned to **semi-supervised** learning
- Initially networks were trained to distinguish modified version of an image from different images
- At the same time, **autoencoders** came back into fashion as they could be trained on unlabelled images
- The architecture that excited a huge following was **generative adversarial networks** for generating images

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Reinforcement Learning

- Reinforcement learning has an ancient history
- However, RL also had a reputation of rarely working
- You need to find a detailed enough representation of the state space that is small enough to exhaustively explore
- DeepMind changed that with various Go and Chess players that trained by self-play
- Magically, coupling reinforcement learning with deep learning representations and training sufficiently long just works

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Multi-modality

- From early days there were attempts to couple vision with text
- Real progress came with CLIP that used a simple model that learned the correlations between parts of images and captions
- This idea has a long history
- But implemented by Open-AI, they did this on 400 million image-caption pairs
- CLIP remains a foundation for a huge number of applications from scene understanding to explainable-AI

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- Real progress came with CLIP that used a simple model that learned the correlations between parts of images and captions
- This idea has a long history
- But implemented by Open-AI, they did this on 400 million image-caption pairs
- CLIP remains a foundation for a huge number of applications from scene understanding to explainable-AI

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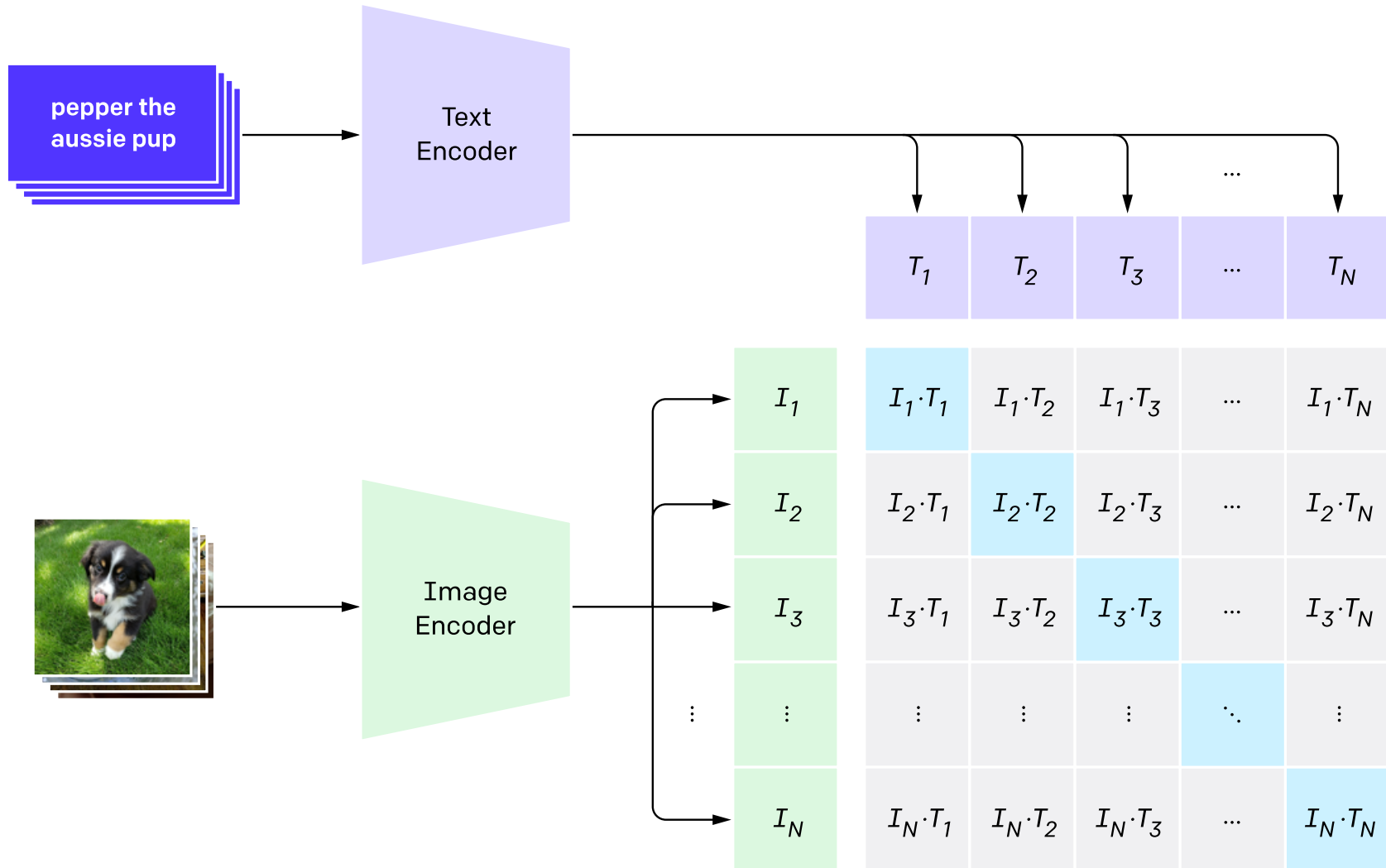
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CLIP

1. Contrastive pre-training



Natural Language Processing

- Natural Language Processing has a history going back to the 1960s
- Neural networks changed the field in the 1980s
- Then Bayesian inference and statistical models took over the field
- With the rise of Deep Learning, first CNNs and then LSTMs started to compete with statistical learning models
- Although better than competing methods, these were still comically bad

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And Then Came Transformers

- The transformer architecture scales to large networks, but trains really well
- Also NLP turns out to be supremely well adapted to semi-supervised learning
- Next word prediction is extraordinary easy to implement and allows you to run on every document you can legally (or illegally) get your hands on
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Attention

Words

The

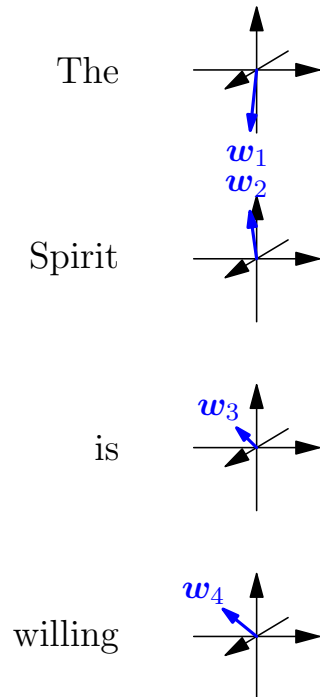
Spirit

is

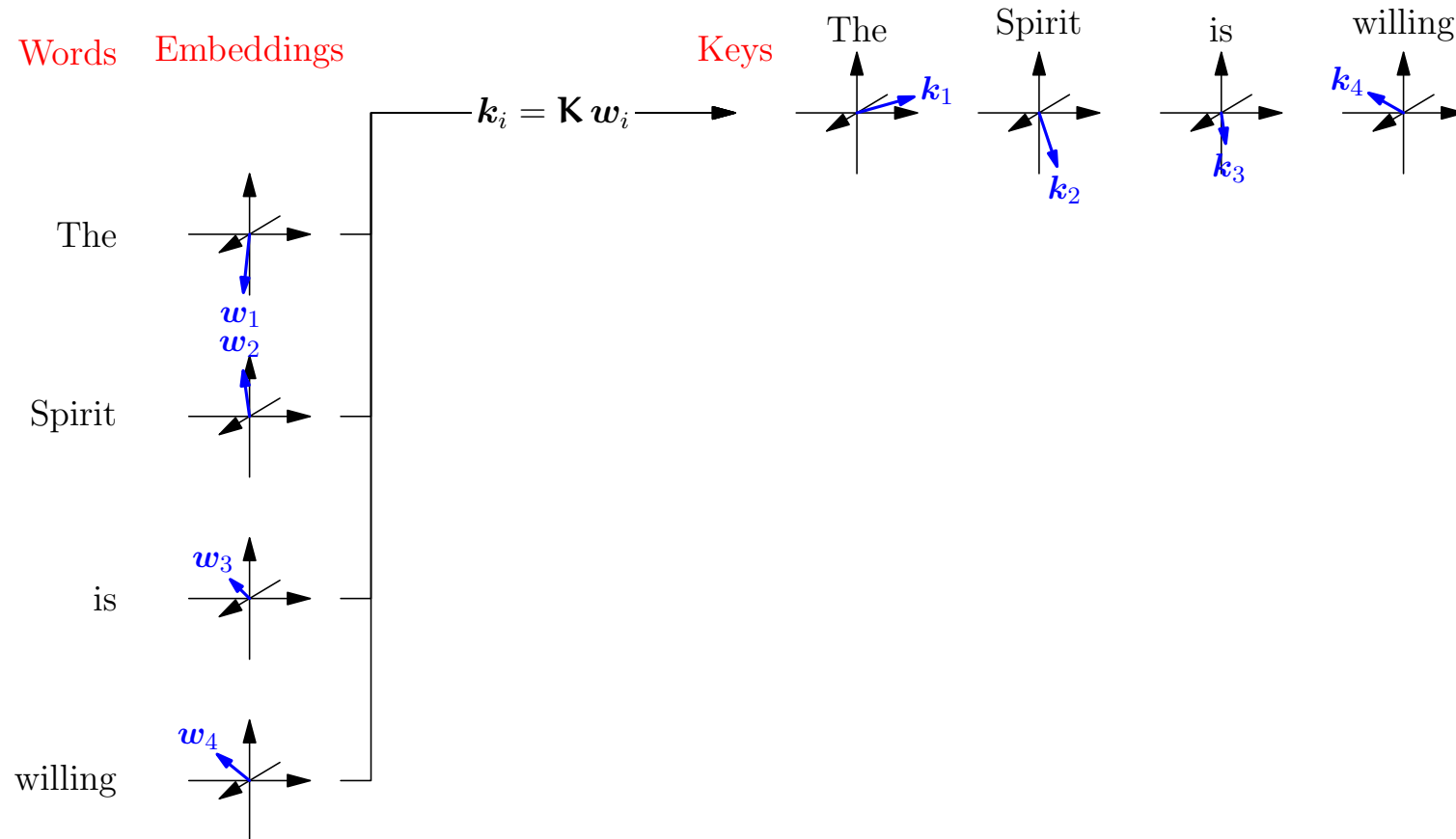
willing

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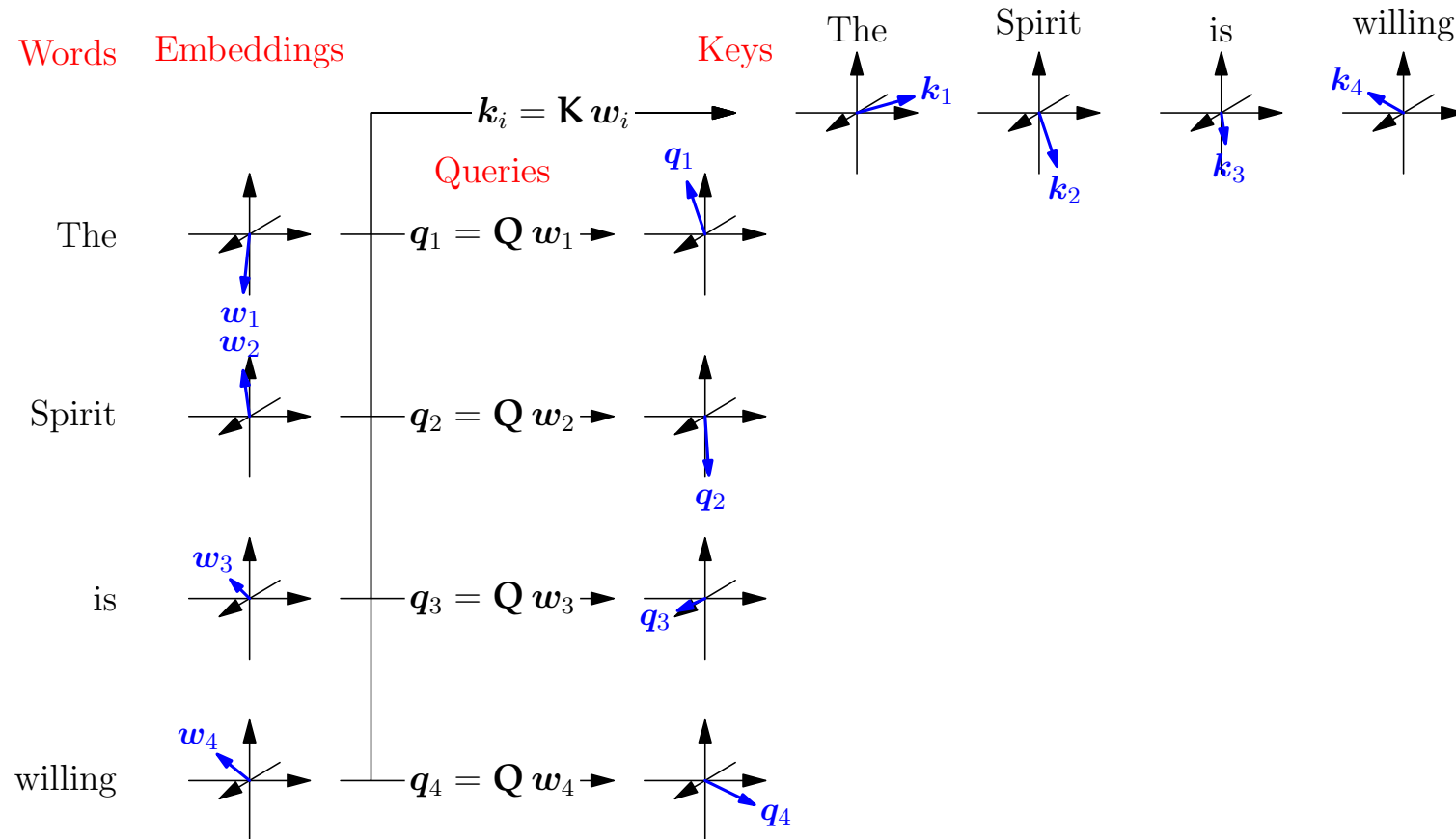
Words Embeddings



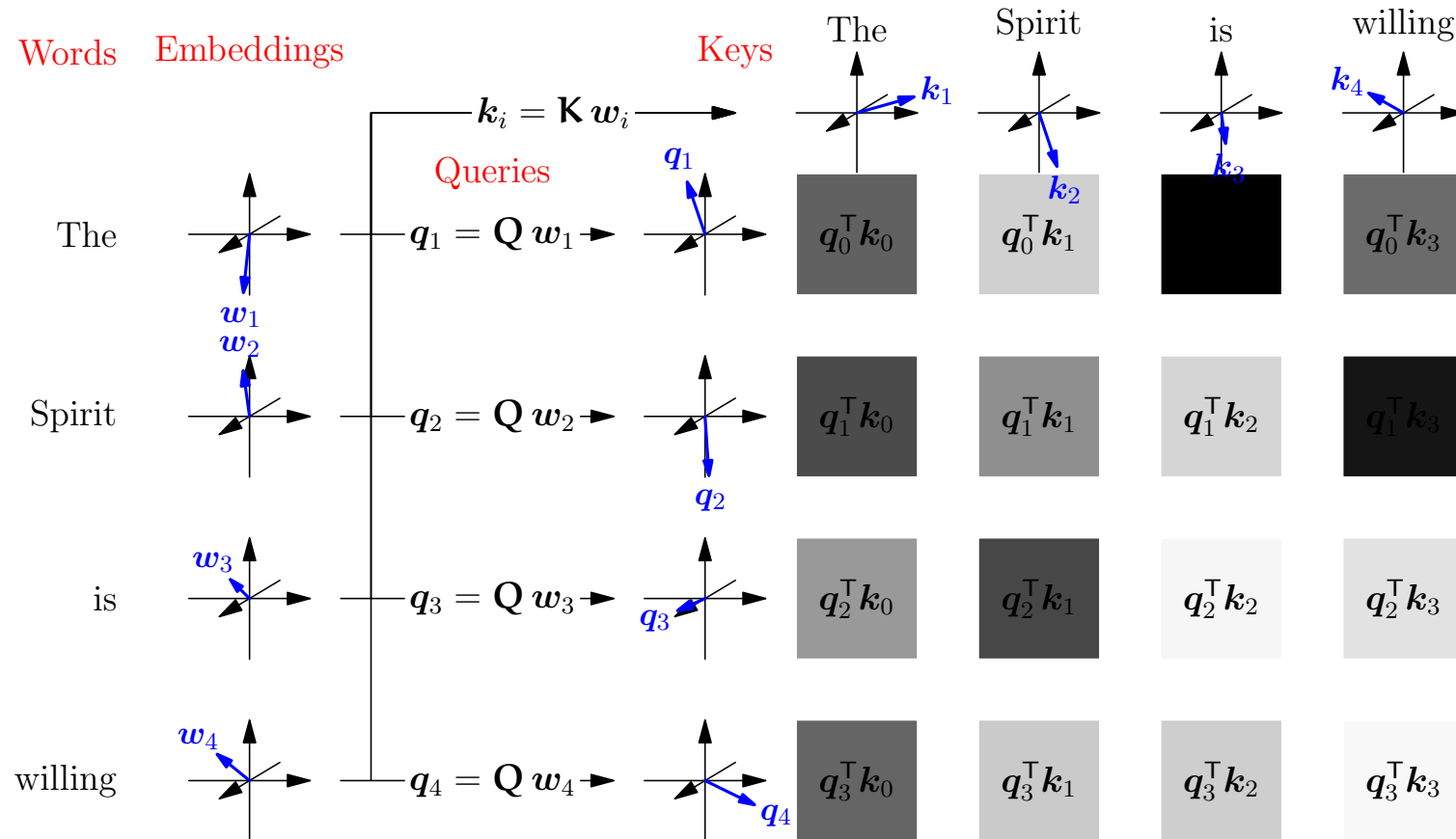
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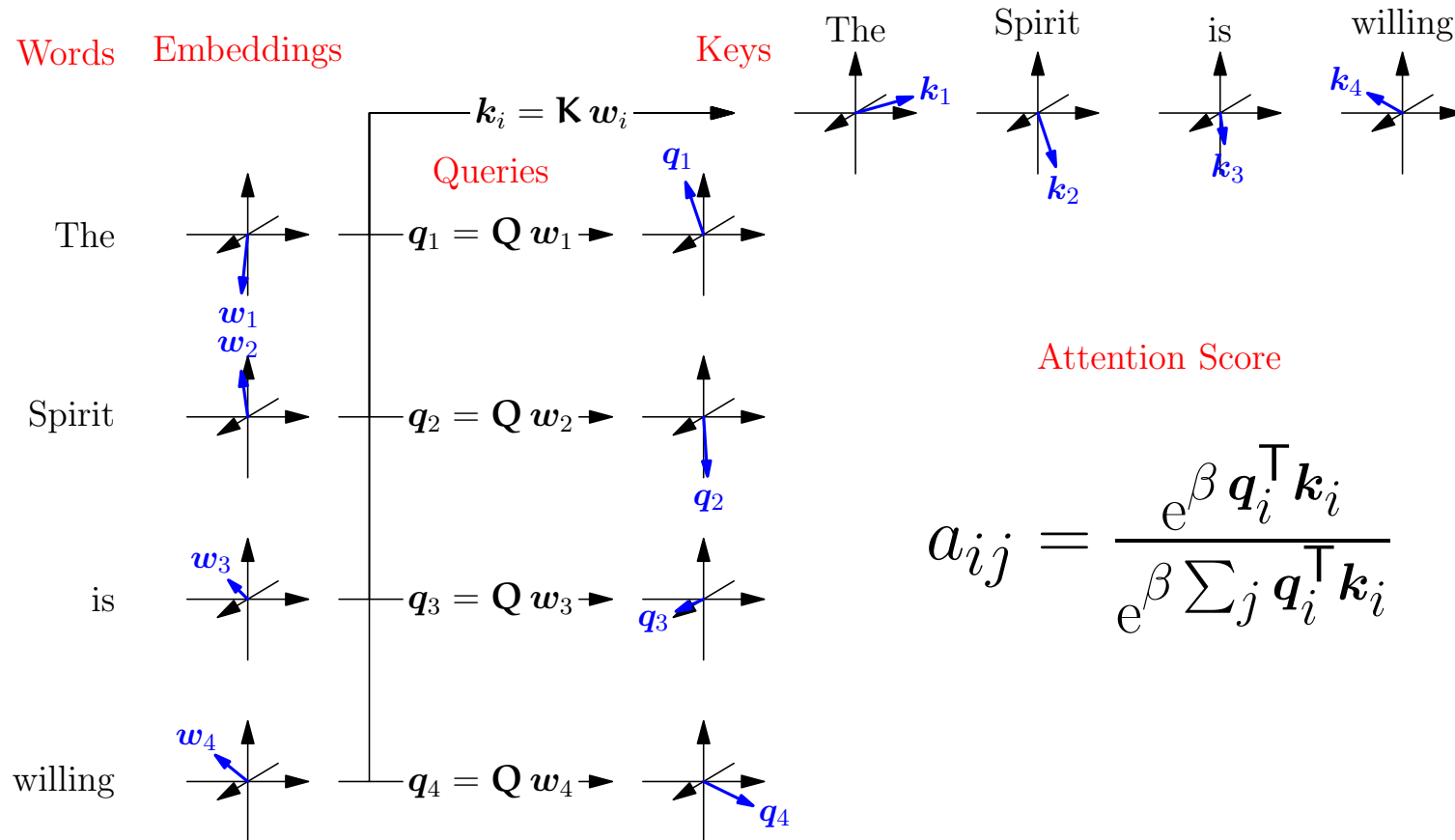
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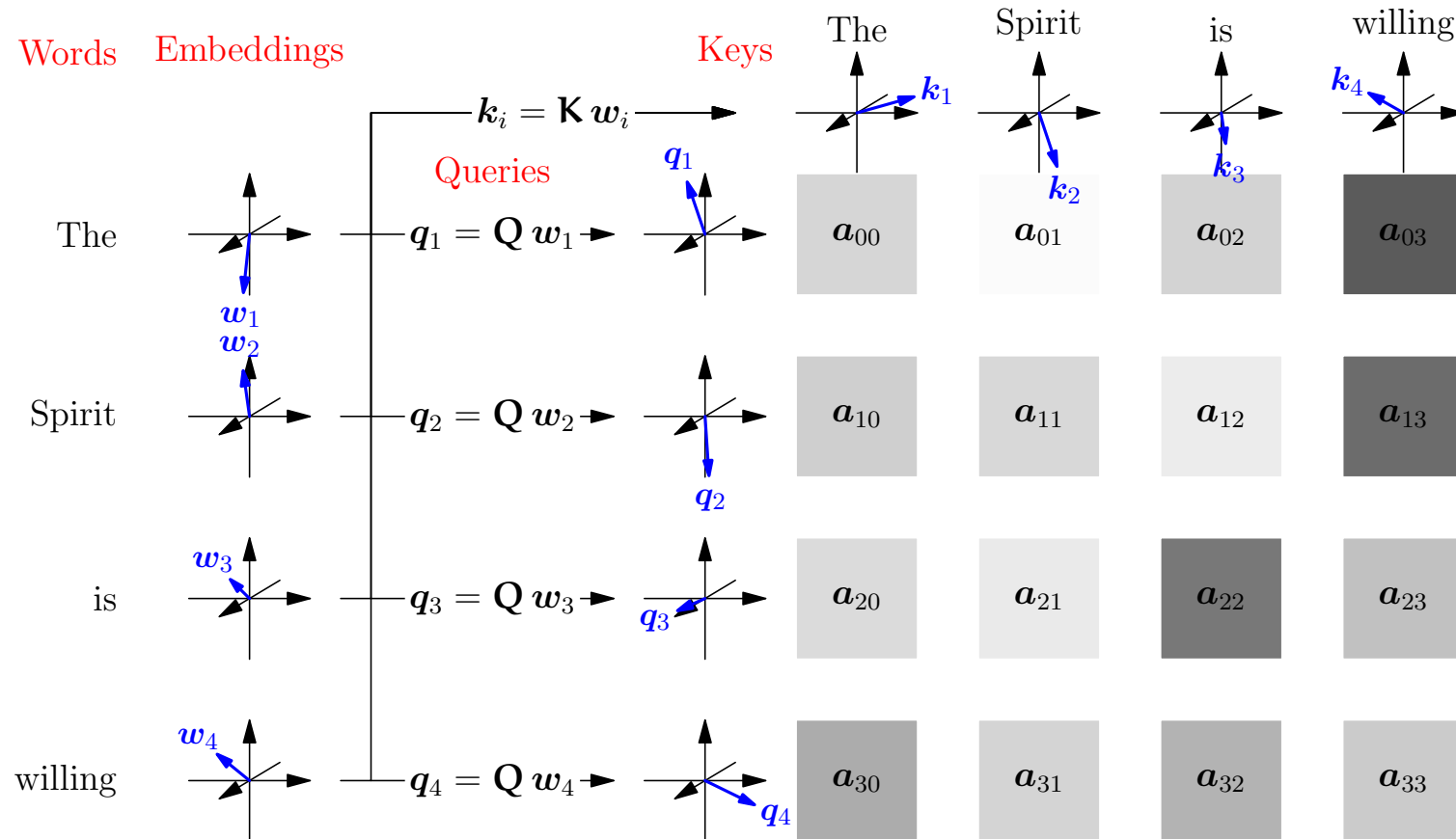
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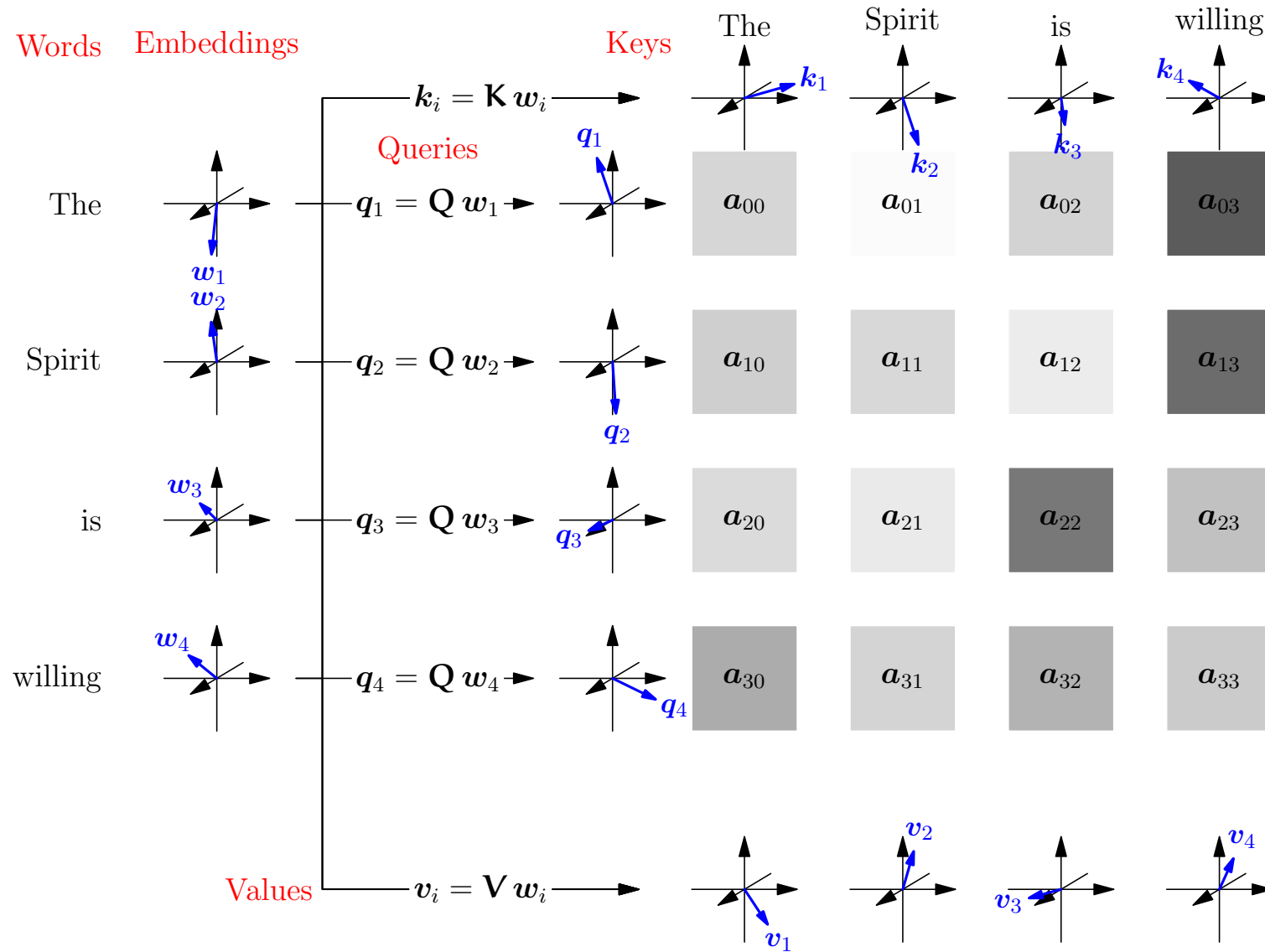
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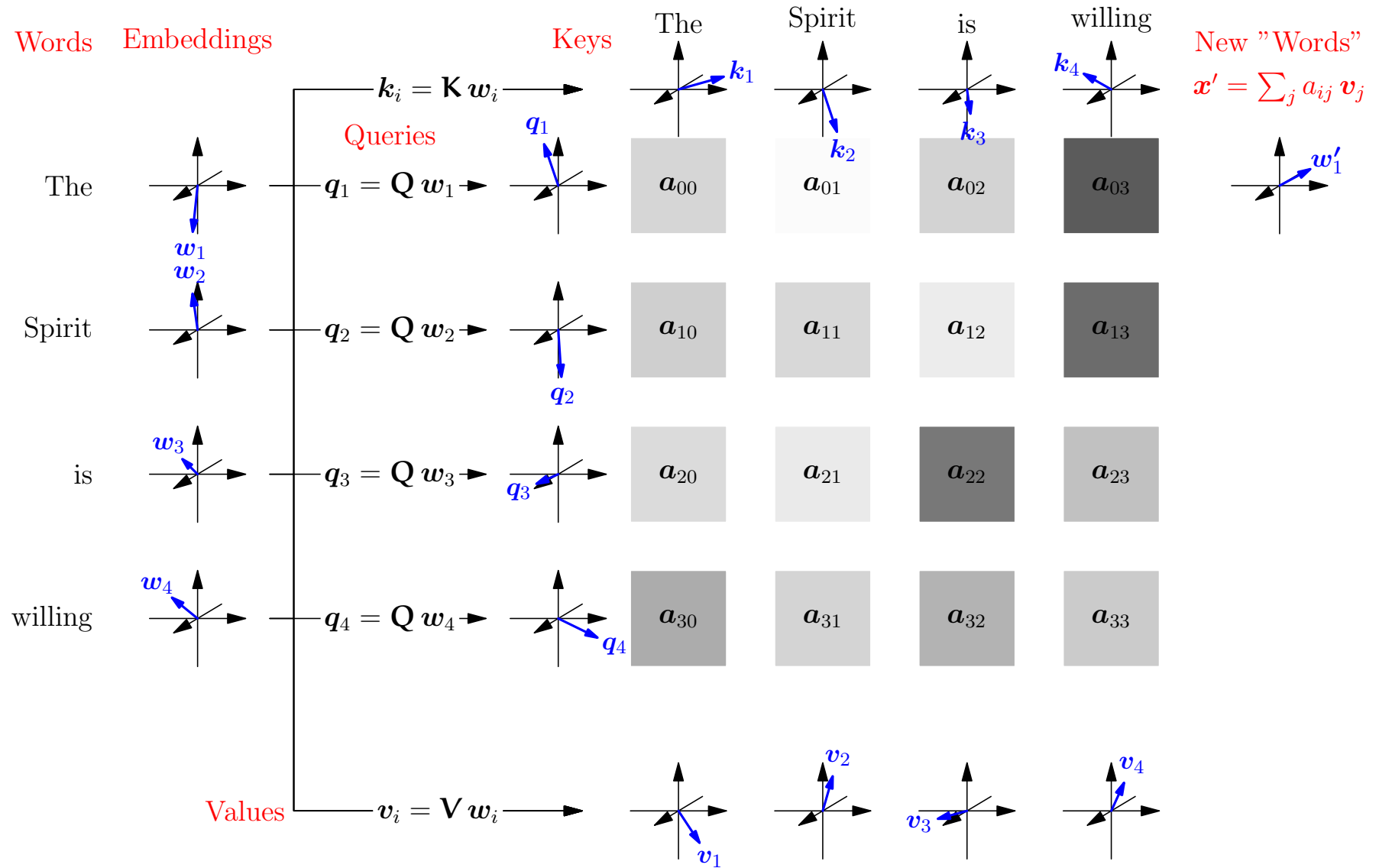
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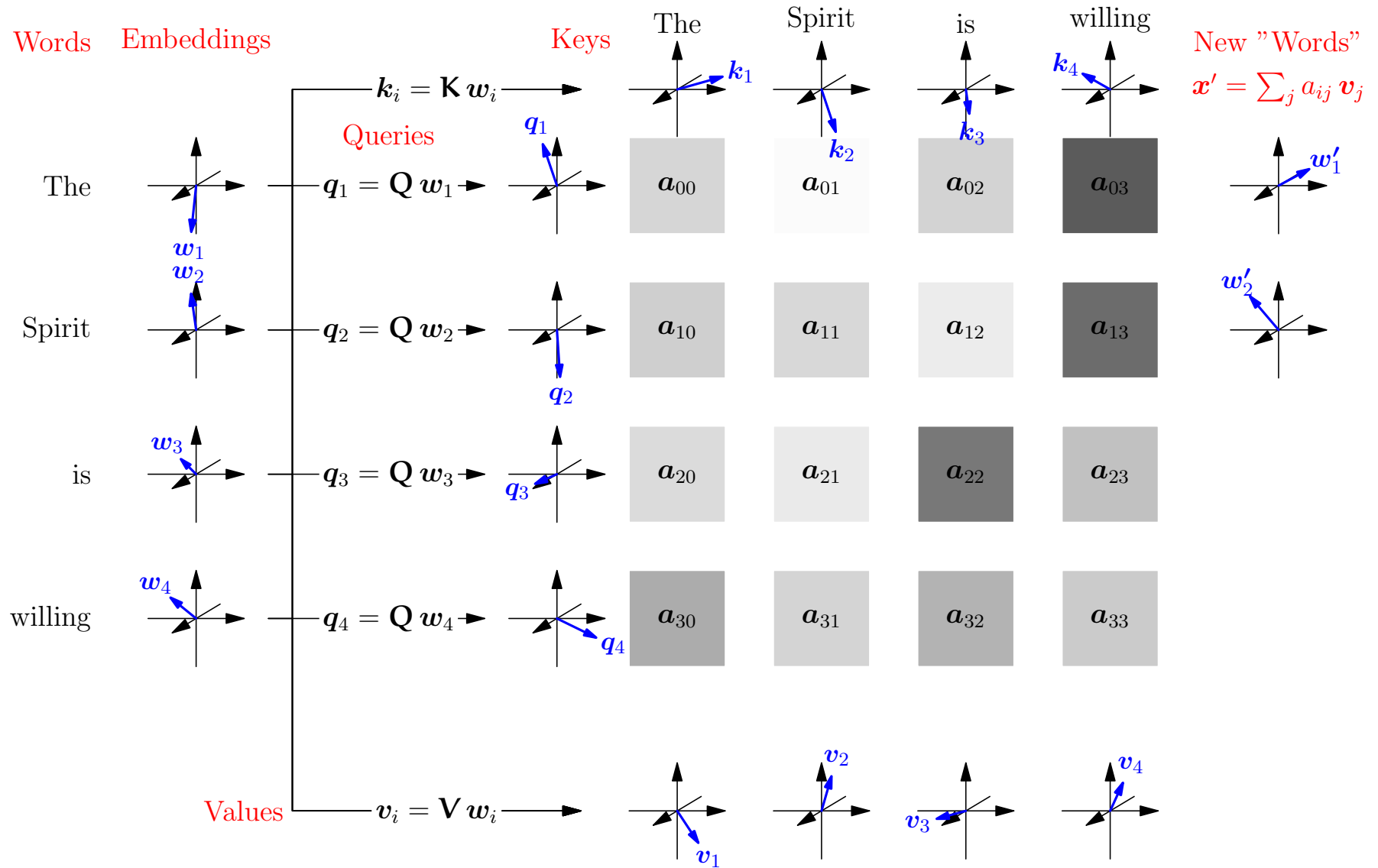
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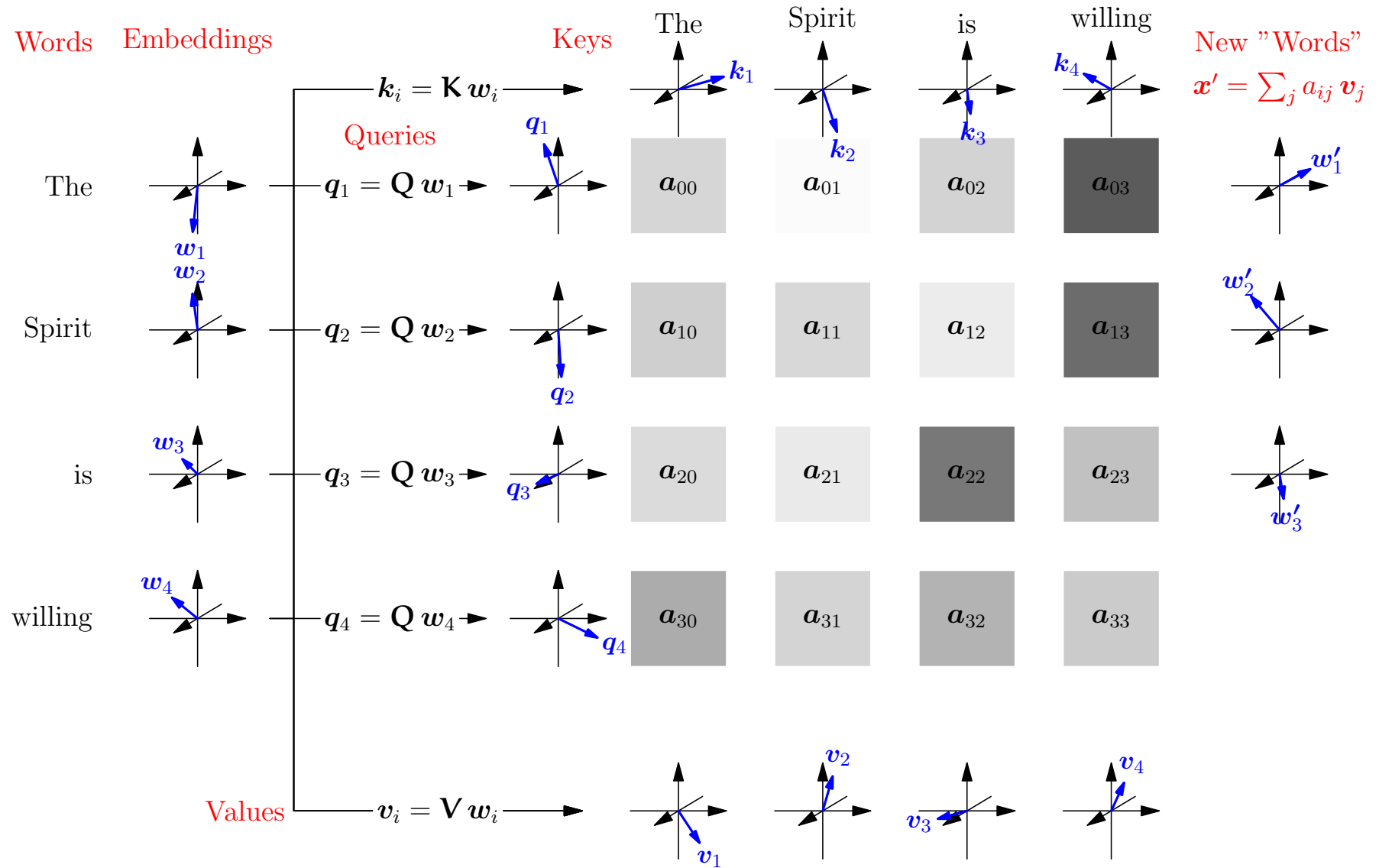
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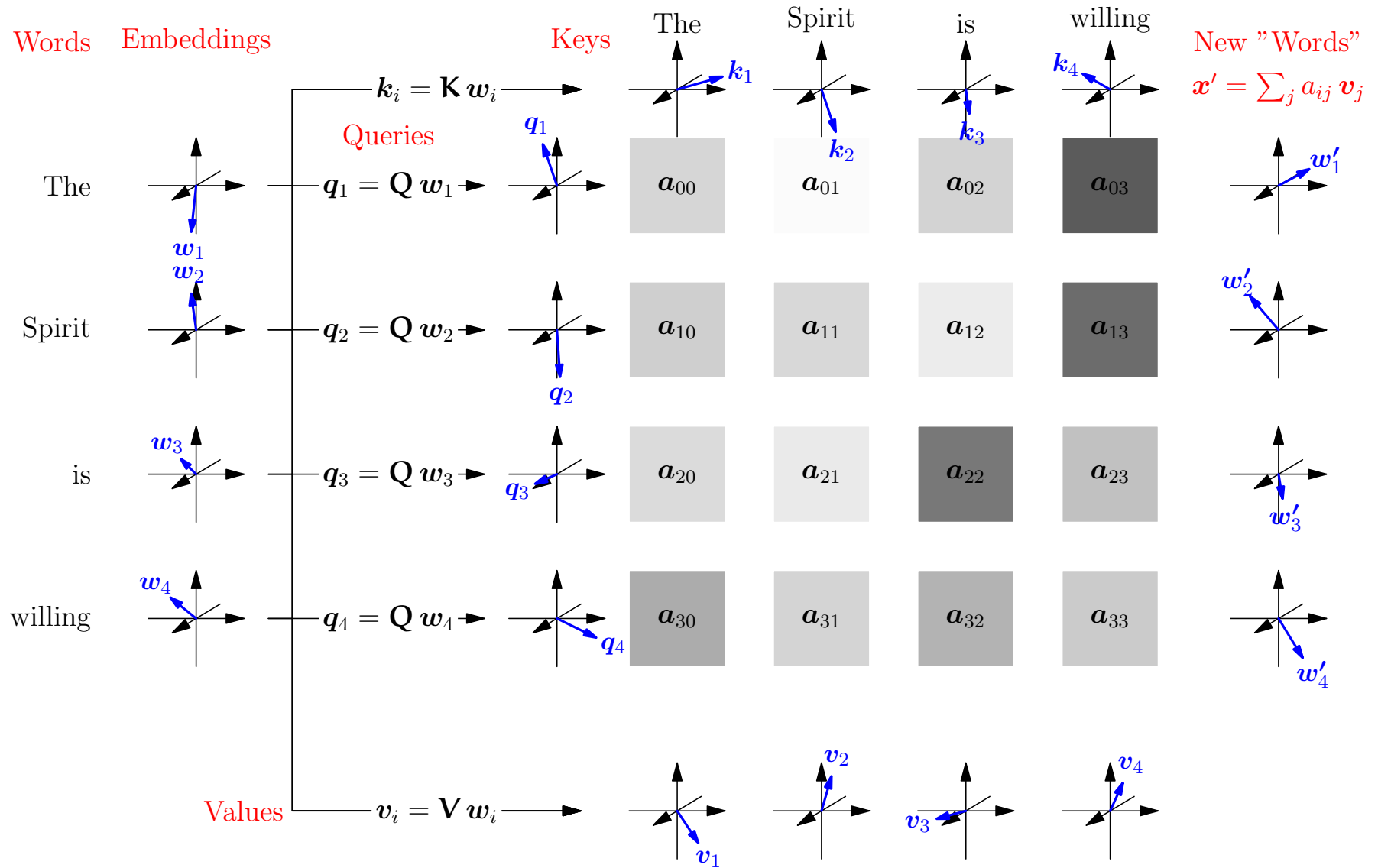
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Training is More Important than Architecture

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- Issues include, embeddings and context window size
- But the real breakthroughs came in the training
- Using huge, well curated datasets, chain-of-thought reasoning, reinforcement learning, etc. makes the real difference

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Transformers Everywhere

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- That is, if you translate the input the output is translated
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- But they have the advantage of understanding combinations of features
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- Today we try to gain the advantages of both architectures

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- Issues around bias, replacing peoples jobs, accountability are all now prescient
- Explainable AI has emerged as an important field

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AI and Big Business

- Because huge models trained on huge datasets now rule the roost, large companies play a disproportionate role in driving AI
- They tend to only work on problems where there is money
- Although, some companies, such as DeepMind, get involve with non-profit prestige projects
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